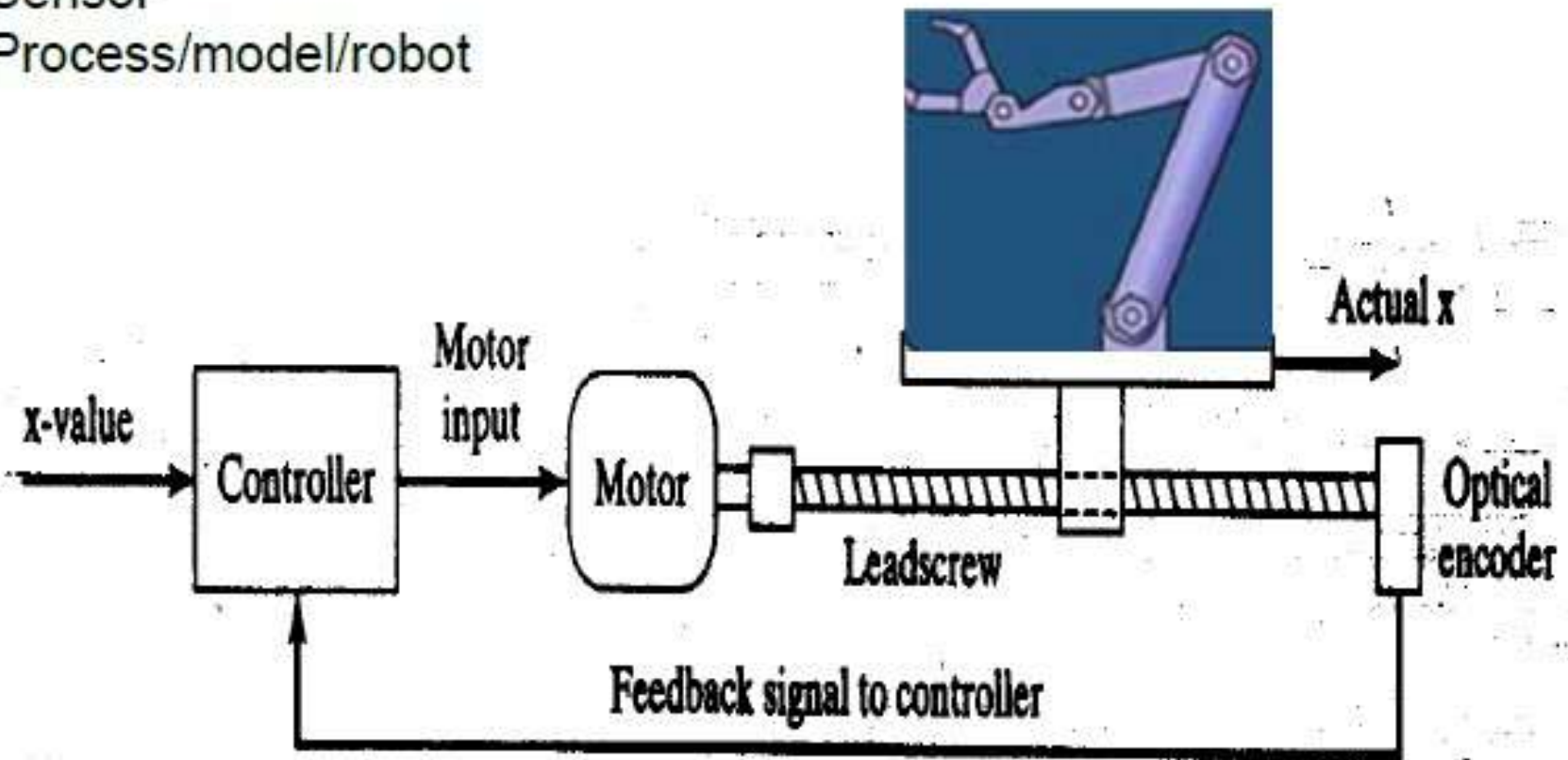


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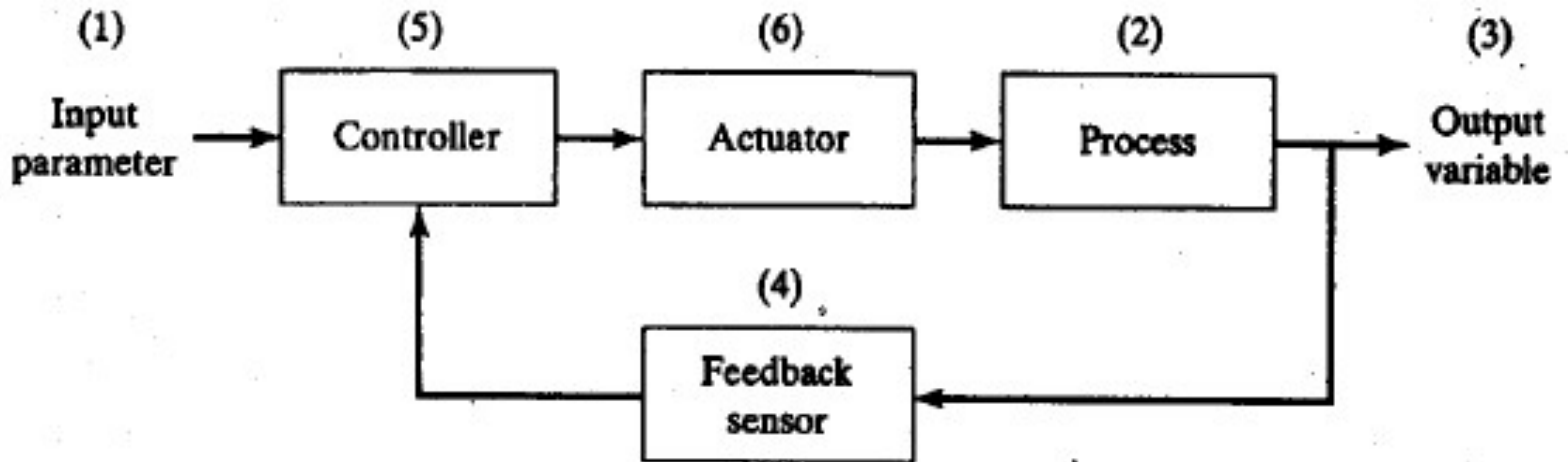
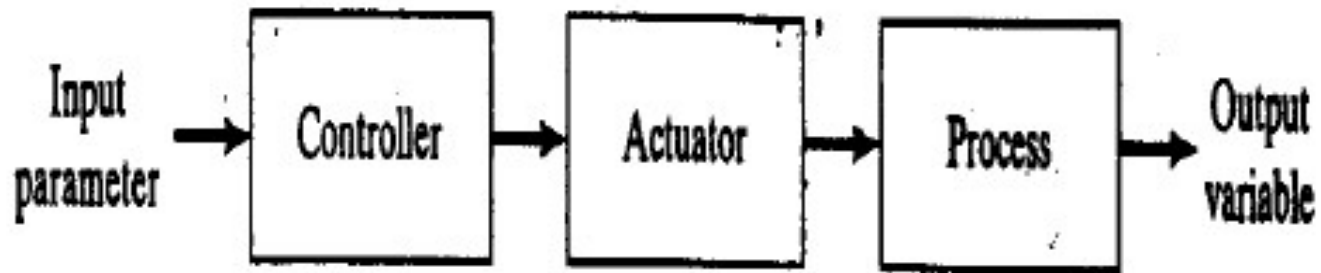
ROBOTICS - ACTUATORS AND SENSORS

Sub-systems in robot control

- Controller
- Actuator
- Sensor
- Process/model/robot



Open loop and closed loop



ACTUATORS FOR ROBOTS:

1. Actuators are used in order to produce mechanical movement in robots.
2. Actuators are the muscles of robots. There are many types of actuators available depending on the load involved. The term load is associated with many factors including force, torque, speed of operation, accuracy, precision and power consumption.

Actuators

- *Electrical* : stepper motors, DC servo motors
- *Pneumatic* : air pressure
- *Hydraulic* : fluid pressure (oil pressure).
- *Advanced actuators* : ultrasonic motors, artificial muscles, molecular motors.

1. Electromechanical actuators convert electrical energy into mechanical energy. Magnetism is the basis of their principle of operation. They are DC, AC and stepper motors.
2. DC motors require a direct current or voltage source as the input signals.
3. AC motors require an alternating current or voltage source

1. Stepper motors have capability of achieving precision angular rotation in both directions and are commonly employed to accommodate digital control technology.
2. Hydraulic and pneumatic actuators are under fluid power actuators. Fluid power refers to energy that is transmitted via a fluid under pressure. When a pressure is applied to a confined chamber containing a piston, the piston will exert a force causing a motion.
3. Materials which undergo some sort of transformations through physical interaction, are referred to as active materials. Piezoelectric (voltage-load), shape-memory alloys (react to heat), magnetostrictive are examples of these materials.

Factors to be considered while choosing the drive system for robots

- a) Accuracy
- b) Repeatability
- c) Degree of freedom
- d) Mobility
- e) Coordinate systems
- f) Gravitational and acceleration force
- g) Backlash, friction and thermal effects
- h) Weight
- i) Power-to-weight ratio
- j) Operating pressure

CHARACTERISTICS OF ACTUATING SYSTEMS

1. Weight, Power-to-weight Ratio, Operating pressure

- 1) Stepper motors are generally heavier than servomotors for the same power.
- 2) The high the voltage of electric motors, the better power-to-weight ratio.
- 3) Pneumatic systems delivers the lowest power-to-weight ratio .
- 4) Hydraulic systems have the highest power-to-weight ratio . In these systems, the weight is actually composed of two portions. One is the hydraulic actuators, and the other is the hydraulic power unit (pump, cylinders, rams, reservoirs, filter, and electric motor). If the power unit must also move with the robot, the total power-to-weight ratio will be much less.

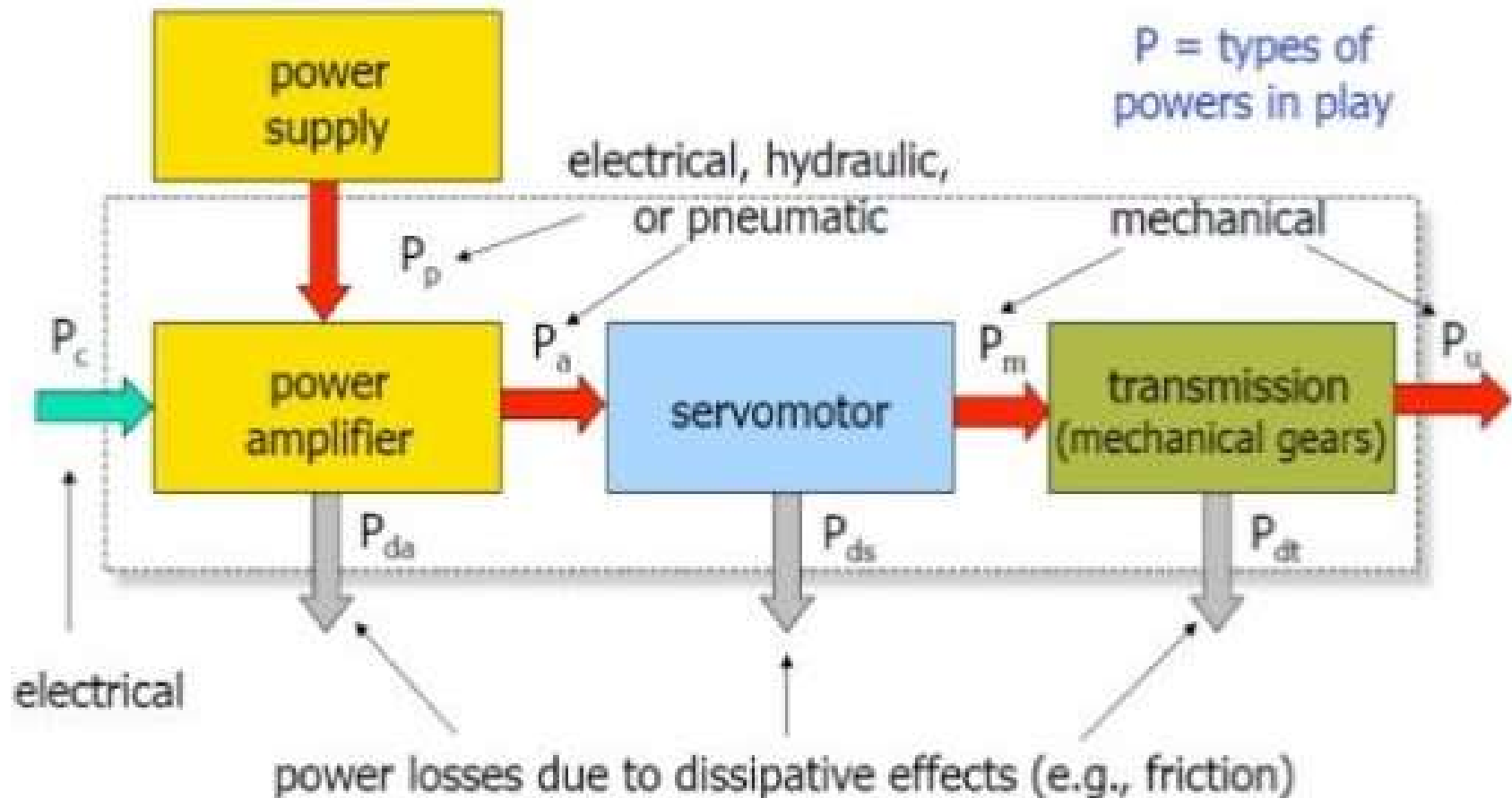
2. Stiffness versus compliance

- 1) Stiffness is the resistance of a material against deformation. The stiffer the system, the larger the load that is needed to deform it. Conversely, the more compliant the system the easier it deforms under the load.
- 2) Stiffness is directly related to the modulus of elasticity of the material. Hydraulic systems are very stiff and non-compliant while pneumatic systems are easily compressed and thus are compliant.
- 3) Stiff systems have a more rapid response to changing loads and pressures and are more accurate.
- 4) Although stiffness causes a more responsive and more accurate systems, it also creates a danger if all things are not always perfect.

Comparison of Actuating Systems

Hydraulic	Electric	Pneumatic
+ Good for large robots and heavy payload + Highest power/weight ratio + Stiff system, high accuracy, better response + No reduction gear needed + Can work in wide range of speeds without difficulty + Can be left in position without any damage — May leak; not fit for clean room applications — Requires pump, reservoir, motor, hoses, and so on — Can be expensive and noisy; requires maintenance — Viscosity of oil changes with temperature — Very susceptible to dirt and other foreign material in oil — Low compliance — High torque, high pressure, large inertia on the actuator	+ Good for all sizes of robots + Better control, good for high precision robots + Higher compliance than hydraulics + Reduction gears reduce inertia on the motor + Does not leak, good for clean room + Reliable, low maintenance + Can be spark-free; good for explosive environments — Low stiffness — Needs reduction gears, increased backlash, cost, weight, and so on — Motor needs braking device when not powered; otherwise, the arm will fall	+ Many components are usually off-the-shelf + Reliable components + No leaks or sparks + Inexpensive and simple + Low pressure compared to hydraulics + Good for on-off applications and for pick and place + Compliant systems — Noisy — Require pressurized air, filter, and so on — Difficult to control their linear position — Deform under load constantly — Very low stiffness, inaccurate response — Lowest power to weight ratio

Actuation systems



HYDRAULIC ACTUATORS

A hydraulic system generally consists of the following parts:

1. Hydraulic linear or rotary cylinders and rams to provide the force or torque needed to move the joints and are controlled by servo valve or manual valve.
2. A hydraulic pump to provide high pressure fluid to the system
3. Electric motor to operate the hydraulic pump.
4. Cooling system to get rid of heat (cooling fans, radiators, and cooled air).
5. Reservoir to keep fluid supply available to the system.
6. Servo valve which is a very sensitive valve that controls the amount and the rate of the fluid to the cylinders. The servo valve is generally driven by a hydraulic servomotor.
7. Sensors to control the motion of the cylinders (position, velocity, magnetic, touch,...)
8. Connecting hoses to transport the pressurized fluid.
9. Safety check valves, holding valves.

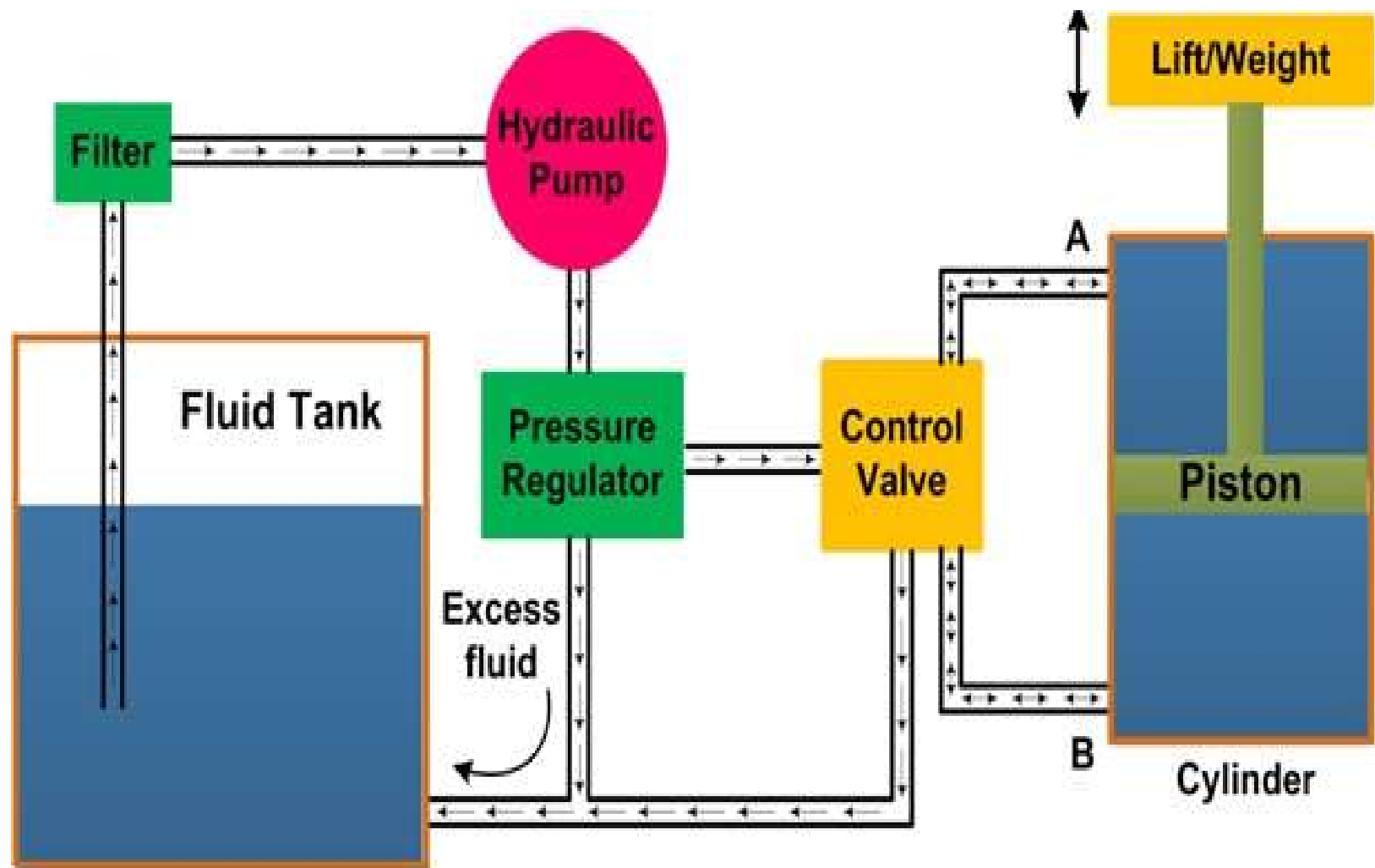


Figure Schematic of hydraulic system

Construction and Working Principle:

- The output shaft transfers the motion or force however all other parts help to control the system.
- The storage/fluid tank is a reservoir for the liquid used as a transmission media.
- The liquid used is generally high density incompressible oil.
- It is filtered to remove dust or any other unwanted particles and then pumped by the hydraulic pump.
- The capacity of pump depends on the hydraulic system design.
- These pumps generally deliver constant volume in each revolution of the pump shaft.
- The pressure regulator is used regulate the pressure of the fluid and also redirects the excess fluid back to the storage tank
- The pressure generated by the hydraulic pump is distributed to the cylinder through pressure regulator and control valves according to the requirement, which is proportional to the amount of load needed to be supported by them.

- The movement of piston is controlled by changing liquid flow from port A and port B.
- The cylinder movement is controlled by using control valve which directs the fluid flow.
- The fluid pressure line is connected to the port B to raise the piston and it is connected to port A to lower down the piston.
- The valve can also stop the fluid flow in any of the port.
- The leak proof piping is also important due to safety, environmental hazards and economical aspects.
- Some accessories such as flow control system, travel limit control, electric motor starter and overload protection may also be used in the hydraulic systems

Advantages and Disadvantages of hydraulic actuators

Advantages:

- Precision motion control over a wide range of speeds and loads
- Robust (longer life and reliability)
- Greater Strength

Disadvantages:

- Higher inertia on robot joints
- High maintenance cost
- Not energy efficient
- Noisy
- Not suited for cleaner environment

Applications:

- Drive spray coating robots
- Material handling robots
- Heavy parts loading and unloading robots
- Gripper mechanisms

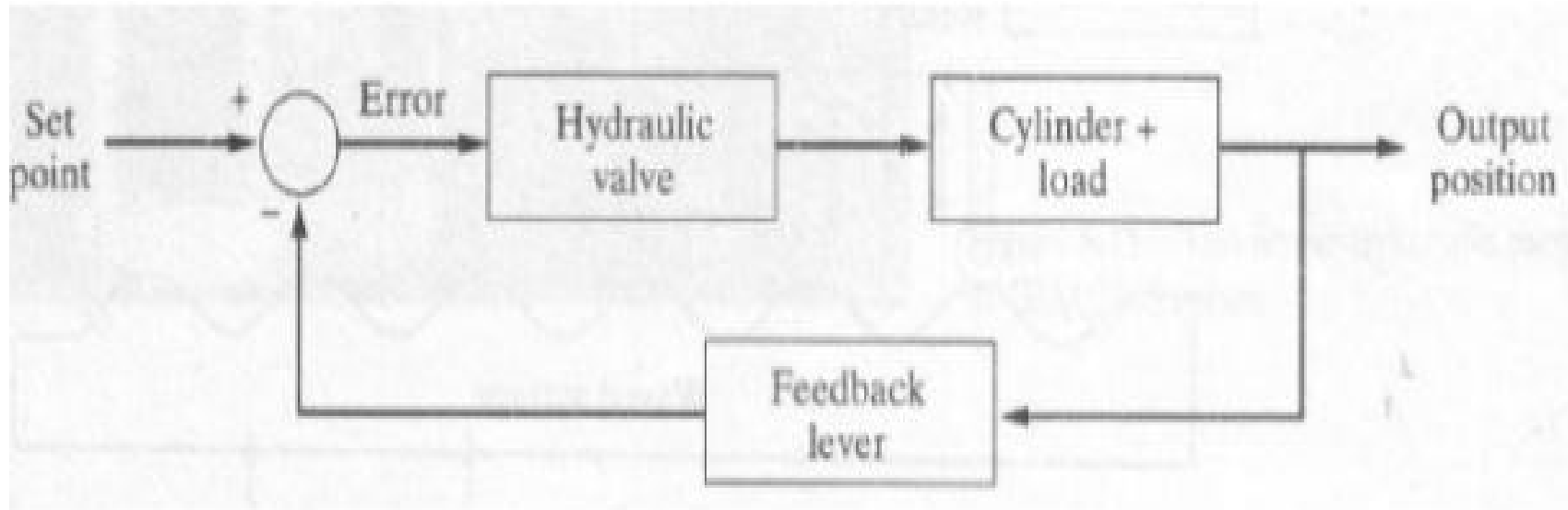


Figure shows the schematic of the, block diagram for the feedback loop.

Pneumatic Actuators

A **pneumatic** control valve actuator converts energy (typically in the form of compressed air) into mechanical motion. These actuators use the principle of Pneumatics (compressed air/gas) to create motion. The motion can be rotary or linear, depending on the type of actuator.

Method of operation:

- Compressed air from the compressor is stored in an air tank and then fed through a pipeline system to the necessary areas of the system.
- A pneumatic actuator (for example, an air cylinder) converts the energy from this compressed air into motion.
- The motion can be rotary or linear, depending on the type of actuator.

TYPES OF PNEUMATICS ACTUATORS

Pneumatic cylinders can be used to get linear, rotary and oscillatory motion. There are three types of pneumatic actuator, they are:

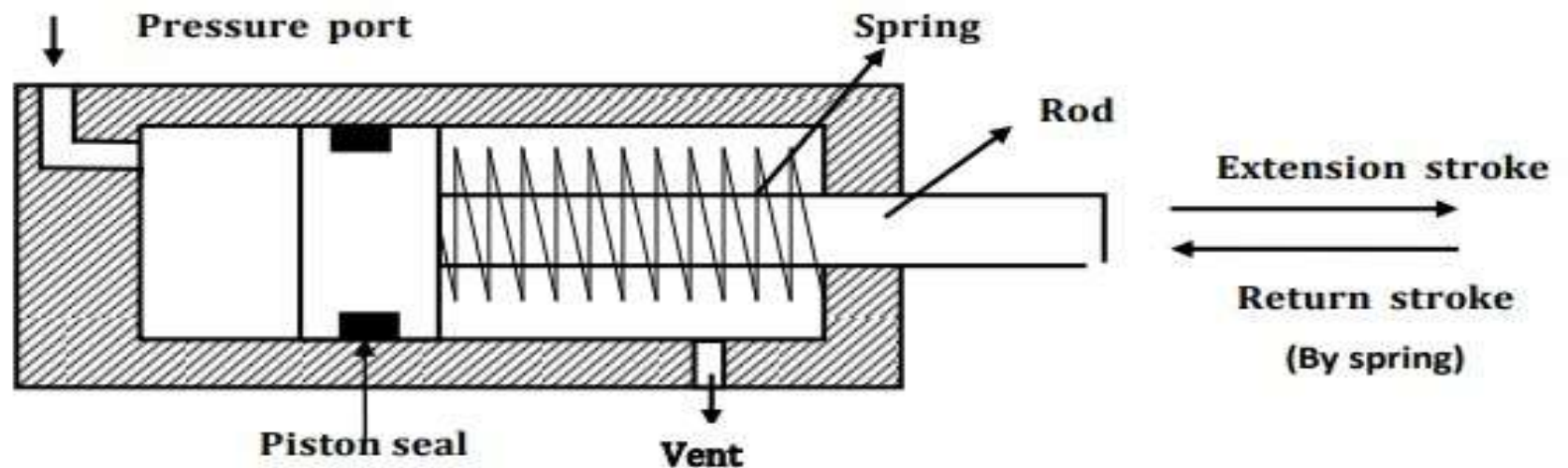
- i) Linear Actuator or Pneumatic cylinders
- ii) Rotary Actuator or Air motors
- iii) Limited angle Actuators

Pneumatic cylinders are devices for converting the air pressure into mechanical force and motion. The pneumatic cylinders are basically used for single purpose application such as **clamping, stamping, ejecting** and many other applications.

Single acting cylinder has one working port. Forward motion of the piston is obtained by supplying compressed air to working port. Return motion of piston is obtained by spring placed on the rod side of the cylinder.

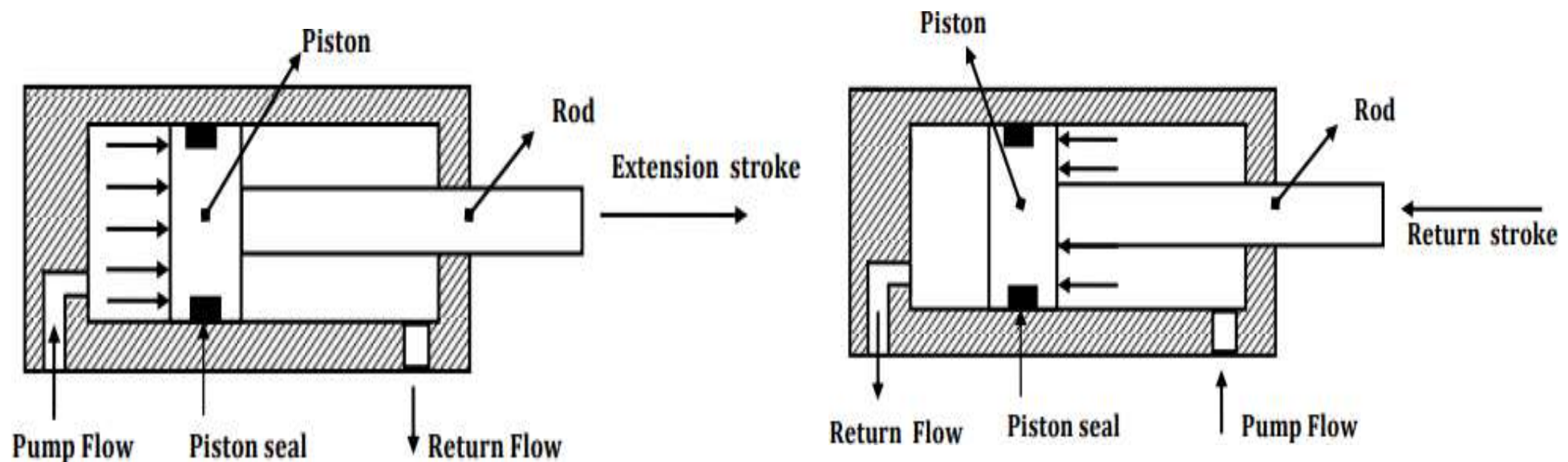
Single acting cylinders are used where force is required to be exerted only in one direction. Such as clamping, feeding, sorting, locking, ejecting, braking etc.,

Single acting cylinder is usually available in short stroke lengths [maximum length up to 80 mm] due to the natural length of the spring. Single Acting Cylinder exert force only in one direction. Single acting cylinders require only about half the air volume consumed by a double acting cylinder for one operating cycle.



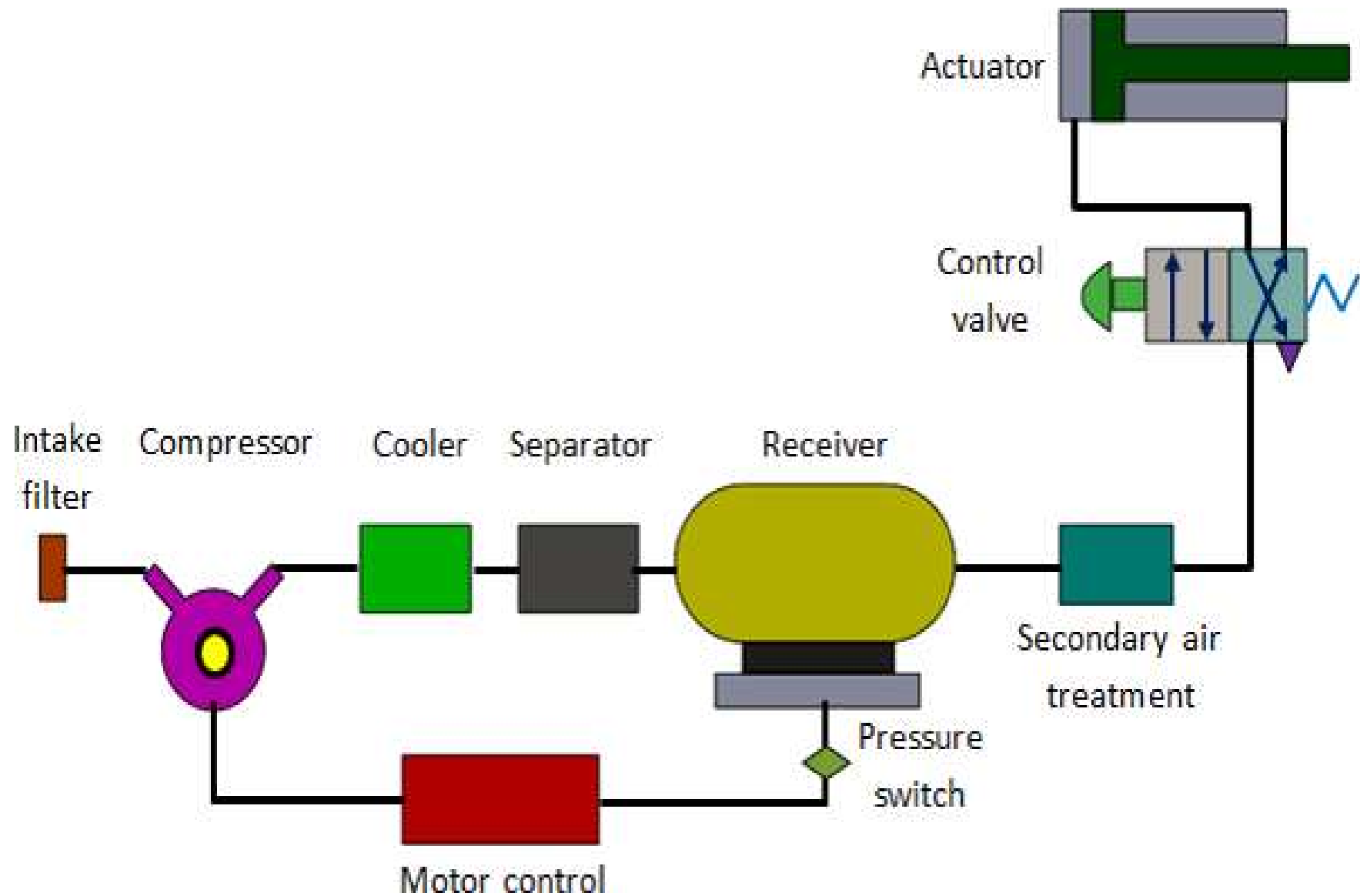
Construction features of single acting cylinder

The construction features of double acting cylinder are shown in **Figure** The construction of double acting cylinder is similar to that of a single cylinder. However, there is no return spring. In double acting cylinder, air pressure can be applied to either side (supply and exhaust) of the piston, thereby providing a pneumatic force in both directions. The double acting cylinders are mostly commonly used in the application where larger stroke length is required.



Double acting cylinder

Basic Components of Pneumatic System



- a. **Air filters:** These are used to filter out the contaminants from the air.
- b. **Compressor:** Compressed air is generated by using air compressors. Air compressors are either diesel or electrically operated. Based on the requirement of compressed air, suitable capacity compressors may be used.
- c. **Air cooler:** During compression operation, air temperature increases. Therefore coolers are used to reduce the temperature of the compressed air.
- d. **Dryer:** The water vapor or moisture in the air is separated from the air by using a dryer.
- e. **Control Valves:** Control valves are used to regulate, control and monitor for control of direction flow, pressure etc.
- f. **Air Actuator:** Air cylinders and motors are used to obtain the required movements of mechanical elements of pneumatic system.
- g. **Electric Motor:** Transforms electrical energy into mechanical energy. It is used to drive the compressor.
- h. **Receiver tank :** The compressed air coming from the compressor is stored in the air receiver.

Advantages

- 1) Higher actuation speed than an electric actuator.
- 2) Actuation speed can be adjusted as desired using a controller.
- 3) Can be used as an emergency shutoff or release valve. (Single acting type; spring return type)
- 4) Can be used for valves that require frequent opening / closing.
- 5) Simple configuration makes it easy to maintain.
- 6) Provide cleaner work environment.

Disadvantages

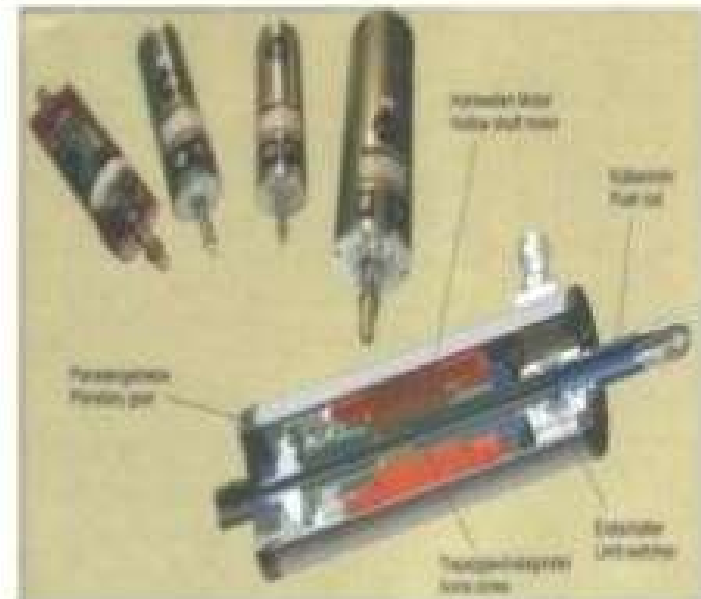
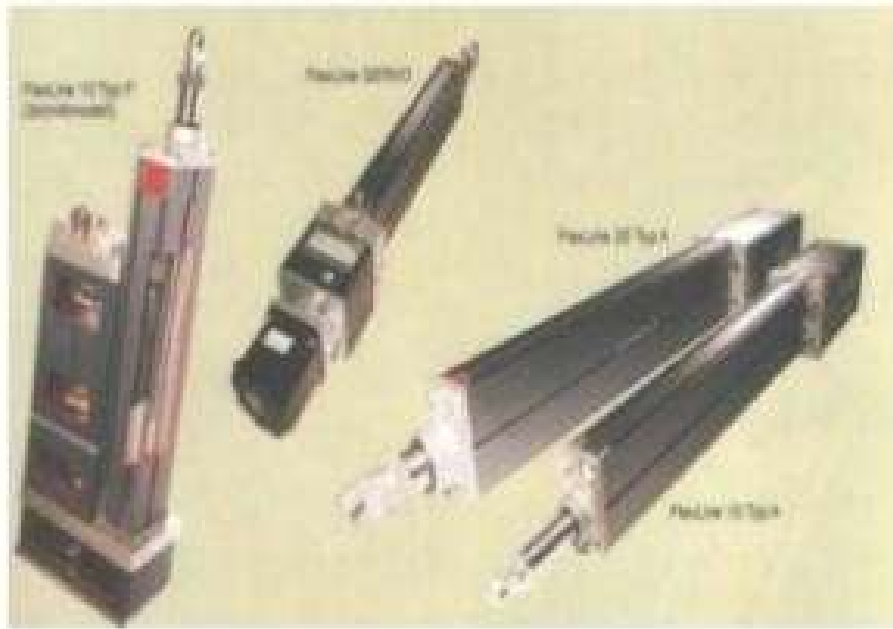
- 1) Additional cost for dust/moisture removing dryer or dust filter is required because air is used.
- 2) Response speed becomes slower (due to the compression of air) where the actuator is significantly distant from the supply air source.
- 3) A larger size actuator is required to obtain high output power.
- 4) Actuation is affected by fluctuation in air pressure and flow rate.

ELECTRICAL ACTUATORS

- 1) Easy to control
- 2) From W to MW
- 3) Normally high velocities 1000 - 10000 rpm
- 4) Several types
- 5) Accurate servo control
- 6) Ideal torque for driving
- 7) Excellent efficiency
- 8) Autonomous power system

ELECTRICAL ACTUATORS

1. Mainly rotating but also linear ones are available
2. Linear movement with gear or with real linear motor



ELECTRICAL ACTUATOR TYPES

1. Servo Motor
2. DC-motors
3. Brushless DC-motors
4. Asynchronous motors
5. Synchronous motors
6. Reluctance motors
7. Stepper Motor

Stepper Motor

- A stepper motor is so named because it is a motor that moves in discrete steps. These motors have a number of coils arranged in phases.
- The power source energizes each phase in sequence, causing the motor to rotate one step for each phase.
- The stepping motion of the motor allows for a great deal of precision and speed control.

Servo Motor

- A servo motor is a closed-loop motor that is linked to an encoder that provides position and speed feedback.
- The mechanism uses this feedback to determine its motion.
- The controller sends an error signal to the motor if it is not in the right position, signalling it to continue its motion until it reaches the desired position.
- The servo motion is suitable for high speed, precision applications.

STEPPER MOTOR	SERVO MOTOR
Open Loop Control	Closed Loop Control
Stepper control systems are optimal for applications that require low-to-medium acceleration, high holding torque, and the flexibility of open or closed loop operation.	Servo control systems are best suited to high speed, high torque applications that involve dynamic load changes.
Typical Steppers motors have typically 50 to 100 poles.	while typical servo motors have only 4 to 12 poles.
Steppers don't require encoders since they can accurately move between their many poles.	whereas servos, with few poles, require an encoder to keep track of their position.
Traditional steppers operate in the open loop constant current mode. This is a cost savings, since no encoder is necessary for most positioning applications.	However, stepper systems operating in a constant current mode creates a significant amount of heat in both the motor and drive.

SERVO SYSTEM

1. Servo is mechanism based on feedback control.
2. The controlled quantity is mechanical.

Properties of Servo Motor

1. High maximum torque/force allows high (de)acceleration
2. High zero speed torque/force
3. High bandwidth provides accurate and fast control
4. Works in all four quadrants
5. Robustness

What is a Servo Motor?

- A servomotor is a rotary actuator or **linear** actuator that allows for precise control of angular or **linear** position, velocity and acceleration.
- It consists of a suitable motor coupled to a sensor for position feedback.
- If you want to rotate an object at some specific angle or distance, then you use servo motor.
- It is just made up of simple motor which run through **servo mechanism**.
- If motor is used is DC powered then it is called DC servo motor, and if it is AC powered motor then it is called AC servo motor.
- The position of a servo motor is decided by electrical pulse and its circuitry is placed beside the motor.
- We can get a very high torque servo motor in a small and light weight packages. Due to these features they are being used in many applications like toy car, Robotics, Machines etc.

TYPES OF SERVO MOTORS

- AC servo motor



- Dc servo motor



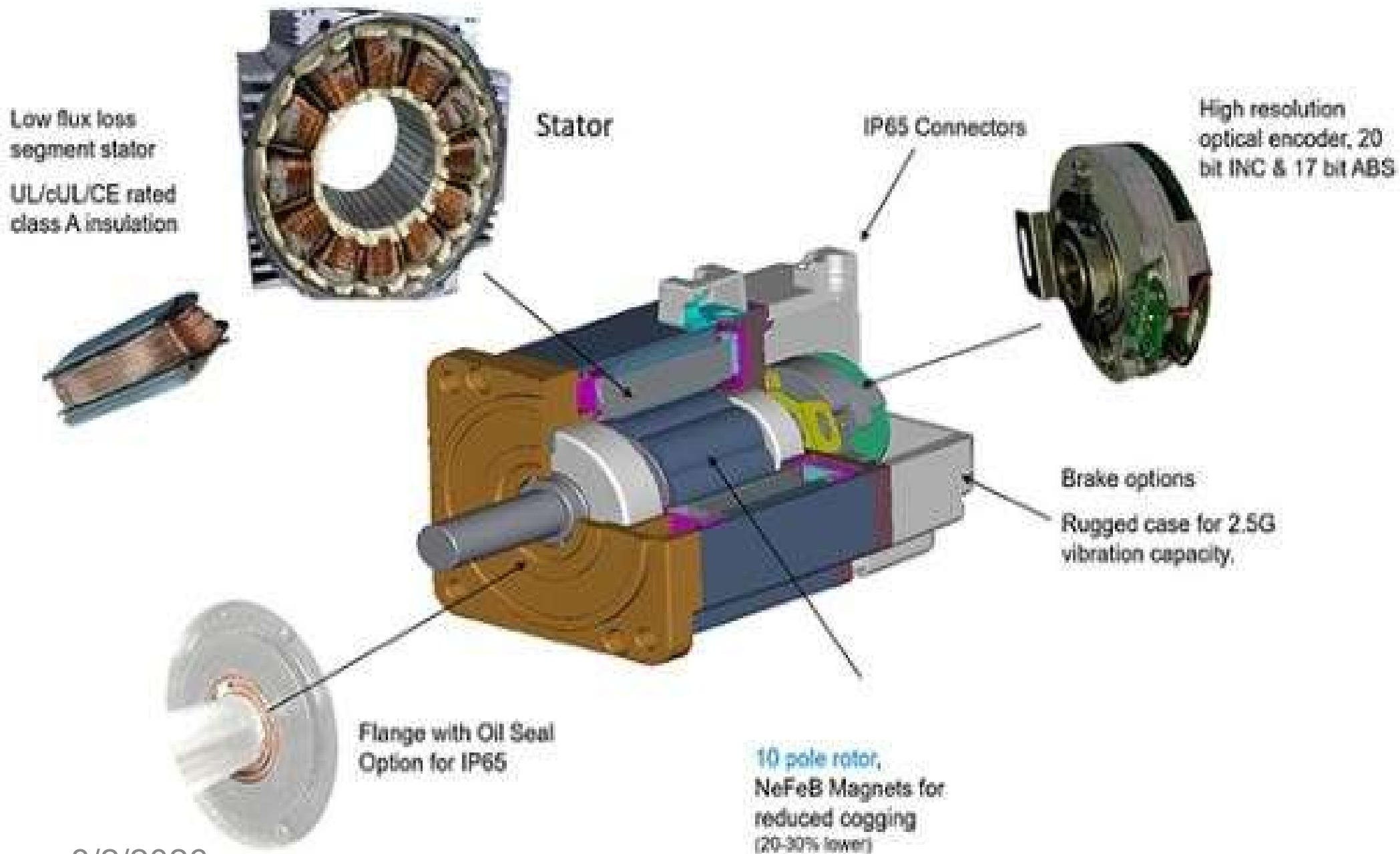
- Continuous rotation servo motor



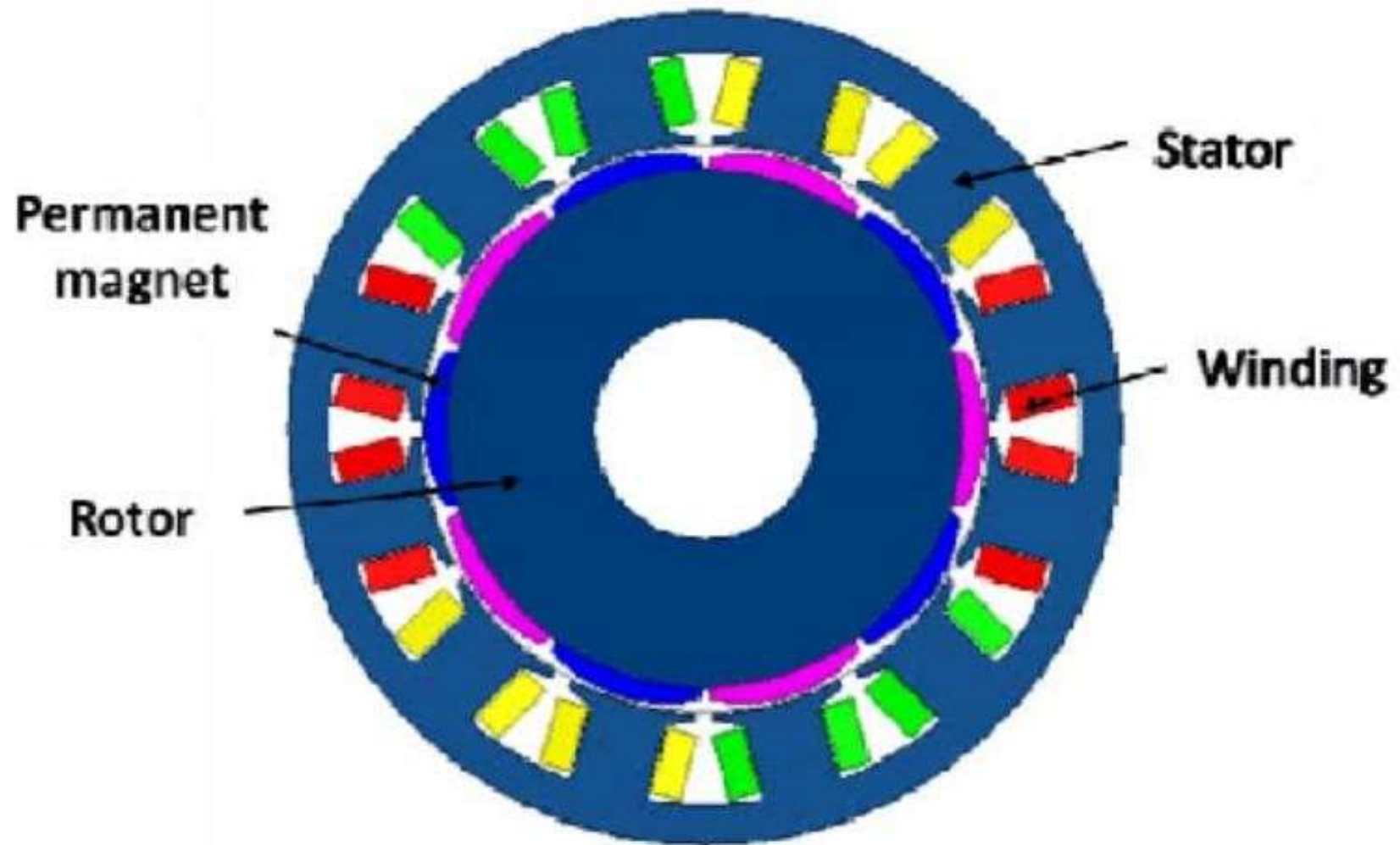
- Linear servo motor



Servo Motor Construction



CONSTRUCTION OF SERVO MOTOR



Stator

From the position of the rotor, a rotating magnetic field is created to efficiently generate torque.

Winding

Current flows in the winding to create a rotating magnetic field.

Bearing

Ball Bearing

Shaft

This part transmits the motor output power. The load is driven through the transfer mechanism (such as the coupling).

Rotor

A high-function rare earth or other permanent magnet is positioned externally to the shaft.

Encoder

The optical encoder always watches the number of rotations and the position of the shaft.

Encoder Cable

Motor Cable



Construction of Servo Motor

Servo motor is Dc motor which consist of following parts:

- Stator winding
- Rotor winding
- Bearing
- Shaft
- Encoder

✓The servo motor consists of a stator and rotor winding.

✓The stator winding is wound on stationary part of the motor and this winding is also called field winding of the motor.

✓The rotor winding is wound on the rotating part of the motor and this winding is also called the armature winding of the motor, made of permanent magnets.

✓The motor consists of two bearing on front and back side for the free movement of shaft.

✓Shaft is basically the iron rod on which the armature winding is coupled. The encoder has the approximate sensor for telling the rotational speed and revolution per minute of the motor. The construction of servo motor is shown in figure.

✓[Working of servo motor](#)

Advantages:

- If a heavy load is placed on the motor, the driver will increase the current to the motor coil as it attempts to rotate the motor.
- High-speed operation is possible.
- Easy maintenance

Disadvantages:

- Since the servomotor tries to rotate according to the command pulses, but lags behind, it is not suitable for precision control of rotation.
- Higher cost.
- When stopped, the motor's rotor continues to move back and forth one pulse, so that it is not suitable if you need to prevent vibration

STEPPER MOTORS

1. A sequence of (3 or more) poles is activated in turn, moving the stator in small “steps”.
2. Very low speed / high angular precision is possible without reduction gearing by using many rotor teeth.
3. Can also perform a “microstep” by activating both coils at once.



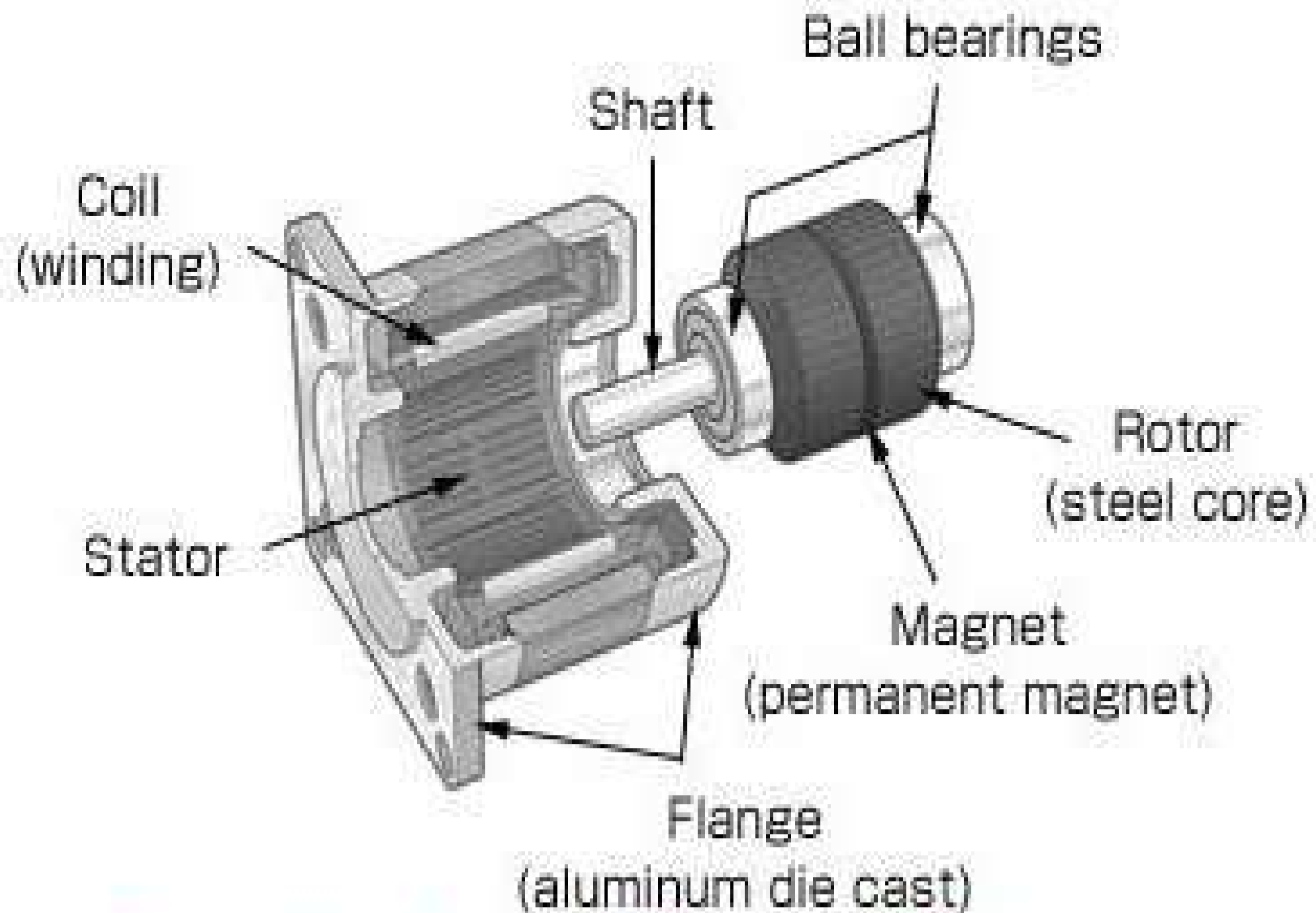
← Stepper Motor

STEPPER MOTOR

1. A Stepper motor is a digital actuator where input is in the form of programmed energization of stator winding and output is in the form of discrete angular position.
2. Rotation of rotor occurs because of magnetic interaction between rotor pole and poles of sequentially energized stator winding.

What is a Stepper Motor?

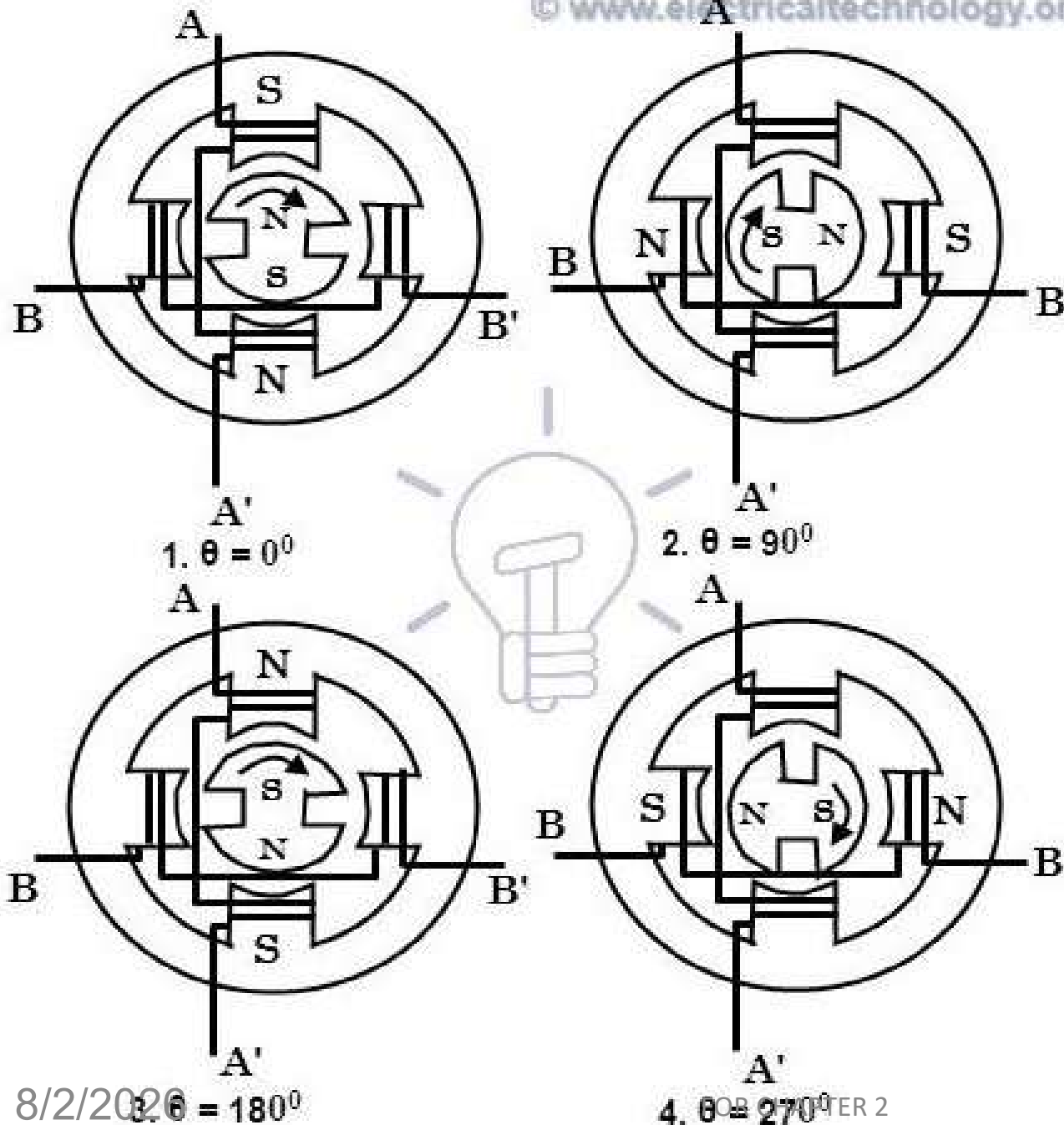
- It is a brushless electromechanical device which converts the train of electric pulses applied at their excitation windings into precisely defined step-by-step mechanical shaft rotation.
- The shaft of the motor rotates through a fixed angle for each discrete pulse. It gets one step movement for a single pulse input.
- When a train of pulses is applied, it gets turned through a certain angle.
- The angle through which the stepper motor shaft turns for each pulse is referred as the step angle, which is generally expressed in degrees.
- Unlike other motors it operates on a programmed discrete control pulses that are applied to the stator windings via an electronic drive.
- The rotation occurs due to the magnetic interaction between poles of sequentially energized stator winding and poles of the rotor.



Construction of Stepper Motor

Working of Permanent Magnet Stepper Motor

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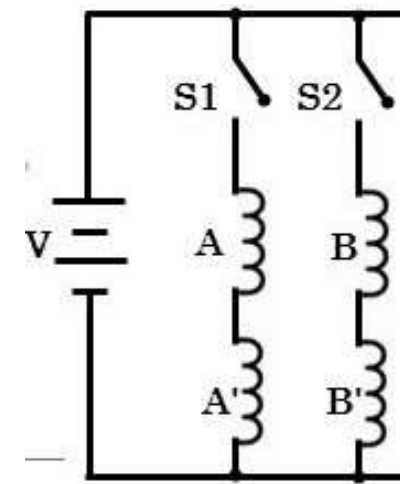
Step angle =

$360 / (N_r \times \text{phase})$

i.e. $360 / (2 \times 2) = 90^\circ$

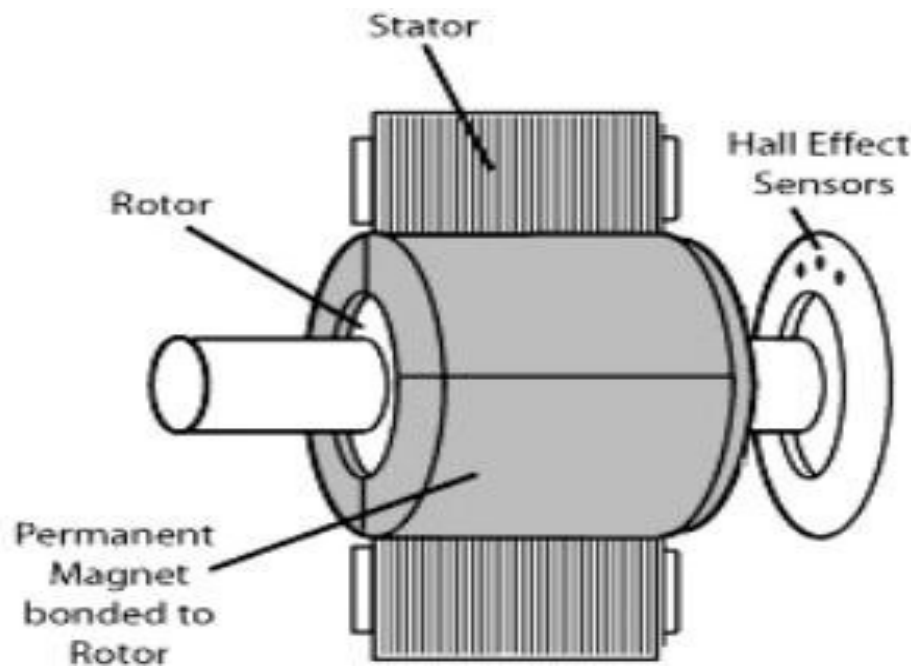
N_r – Number of poles of rotor.

Phase – 2 or 3



What is Brushless DC motor?

Unlike conventional brushed type DC motor, wherein the brushes make the mechanical contact with commutator on the rotor so as to form an electric path between a DC electric source and rotor armature windings, BLDC motor employs electrical commutation with permanent magnet rotor and a stator with a sequence of coils. In this motor, permanent magnet (or field poles) rotates and current carrying conductors are fixed.



A Hall effect sensor (or simply Hall sensor) is a **type of sensor which detects the presence and magnitude of a magnetic field using the Hall effect**. It is named for the American physicist Edwin Hall. Hall sensors are used for **proximity sensing, positioning, speed detection**, and current sensing applications.

Principle of Working – BLDC MOTOR

- The principles for the working of BLDC motors are the same as for a brushed DC motor, i.e., the internal shaft position feedback.
- In the case of a brushed DC motor, feedback is implemented using a mechanical commutator and brushes. Within BLDC motor, it is achieved using multiple feedback sensors.
- In BLDC motors we mostly use a Hall-effect sensor, whenever rotor magnetic poles pass near the hall sensor, they generate a HIGH or LOW-level signal, which can be used to determine the position of the shaft.
- If the direction of the magnetic field is reversed, the voltage developed will reverse too.

Advantages and Disadvantages of Electrical actuators

Advantages:

1. High power conversion efficiency
2. No pollution of working environment
3. They are easily maintained and repaired
4. Light weight
5. The drive system is well suited for electronic control

Disadvantages:

1. Poor dynamic response
2. Conventional gear driven create backlash
3. A larger and heavier motor must be used which must be costly.

OTHER TYPES OF ACTUATORS.

1. Piezoelectric
2. Magnetic
3. Ultrasound
4. Shape Memory Alloys



Advanced actuators: small, low power consumption, micro motion

- **Ultrasonic motors** : micro robots, cameras, micro motion devices ..
- **Artificial muscles** : prosthetic, bio applications..
- **Molecular motors** : bio applications

Ultrasonic motors

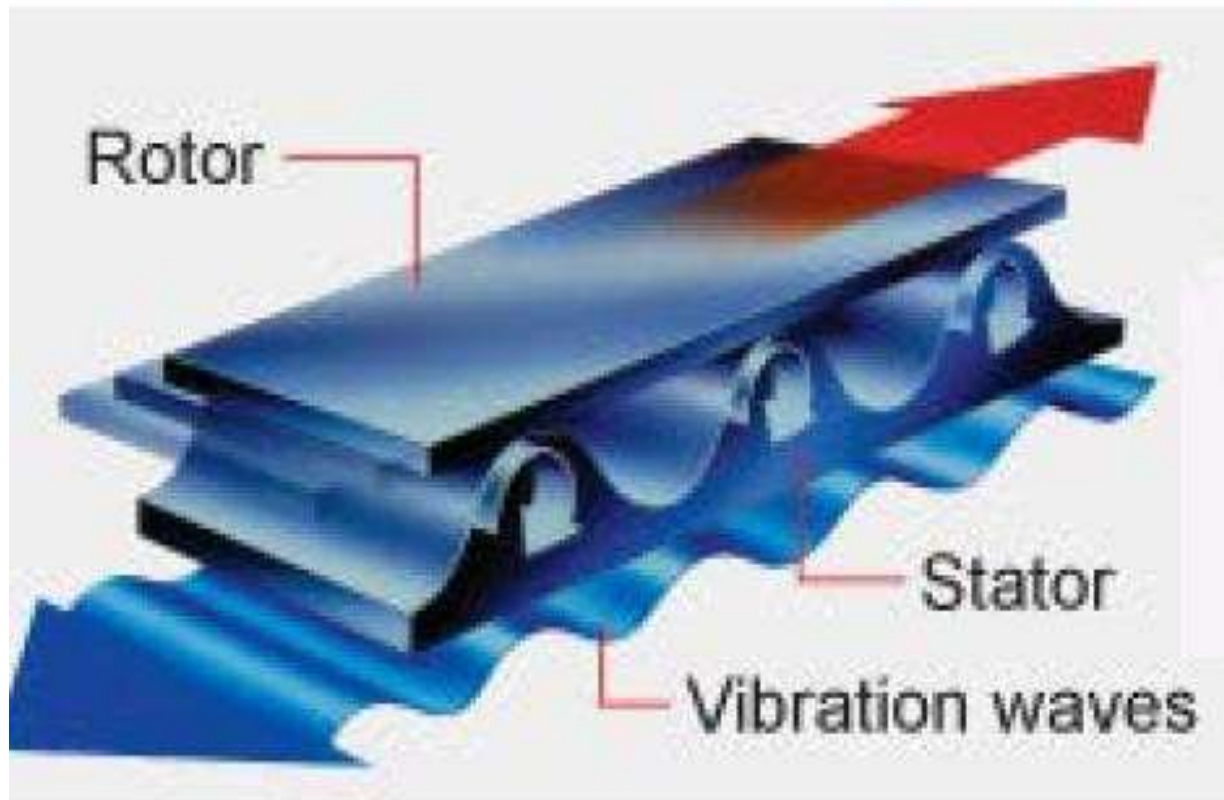


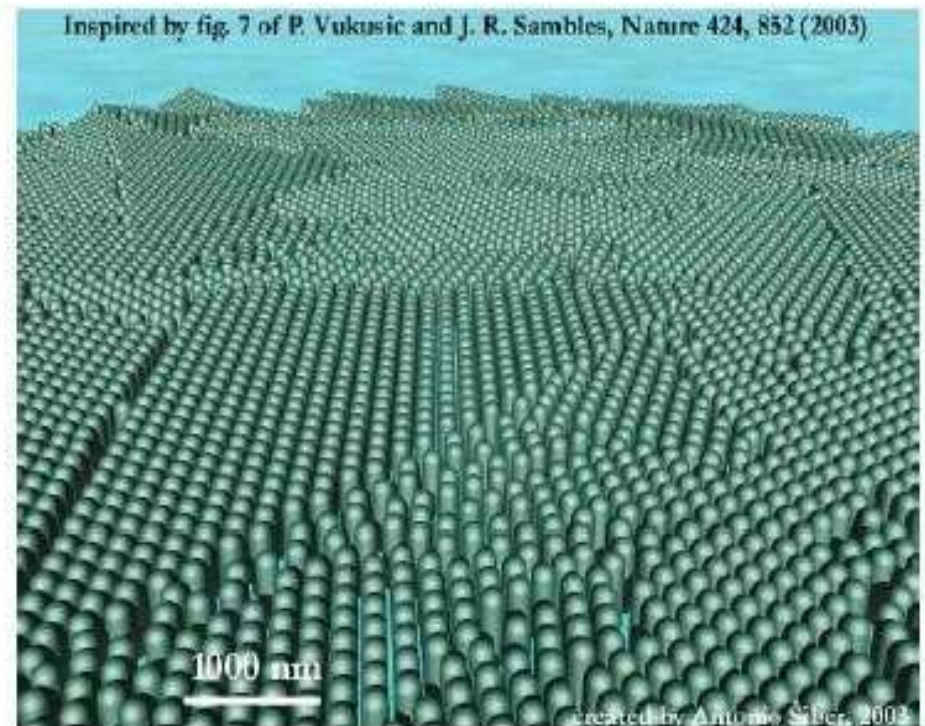
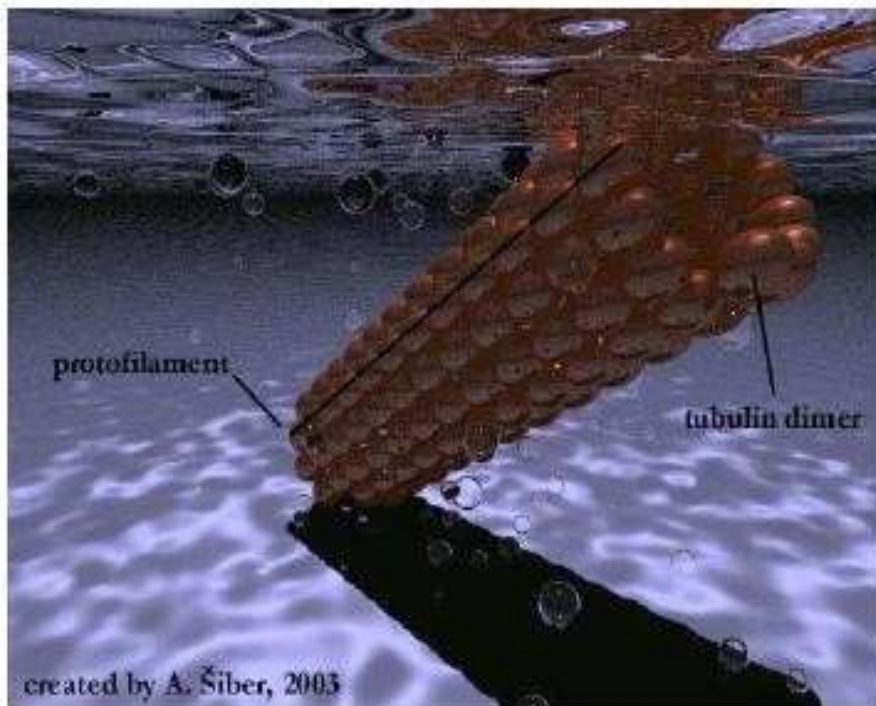
Fig. Motion due to dry friction and vibration.



Fig. Ring motors used in cameras.

Molecular motors

Protein-based molecular motors harness the chemical free energy released by the hydrolysis of ATP in order to perform mechanical work.



Introduction

- **Robotic sensing** is a branch of robotics science intended to give robots sensing capabilities, so that robots are more human-like.
- Robotic sensing mainly gives robots the ability to see, touch, hear and move and uses algorithms that require environmental feedback.
- The use of sensors in robots has taken them into the next level of creativity. Most importantly, the sensors have increased the performance of robots to a large extent. It also allows the robots to perform several functions like a human being.

Robotic sensors are used to estimate a **robot's** condition and environment. These signals are passed to a controller to enable appropriate behaviour. **Sensors** in **robots** are based on the functions of human sensory organs.

General Classification of Sensors

- **Internal sensors:** required for basic working of the system (e.g. position, velocity,).
- **External sensors:** interaction with the environment (vision, force, ...).

There are a number of ways in which sensing devices may be classified:

- **By their type of operation** - analog or digital.
- **Whether the quantity is sensed directly or indirectly** – contact or non-contact
- **By the medium by which they operate** - optical, electrical etc.
- **By their application** – velocity, proximity, position, force etc..

Sensors used for closed loop position control: Internal sensors

- **Position**
- **Velocity**
- **Acceleration**

e.g. potentiometers, encoders, LVDT, Tachometers, Accelerometers

Sensors for interaction with the environment: External sensors

- Touch
- Force
- Pressure
- Slip
- Proximity
- Vision

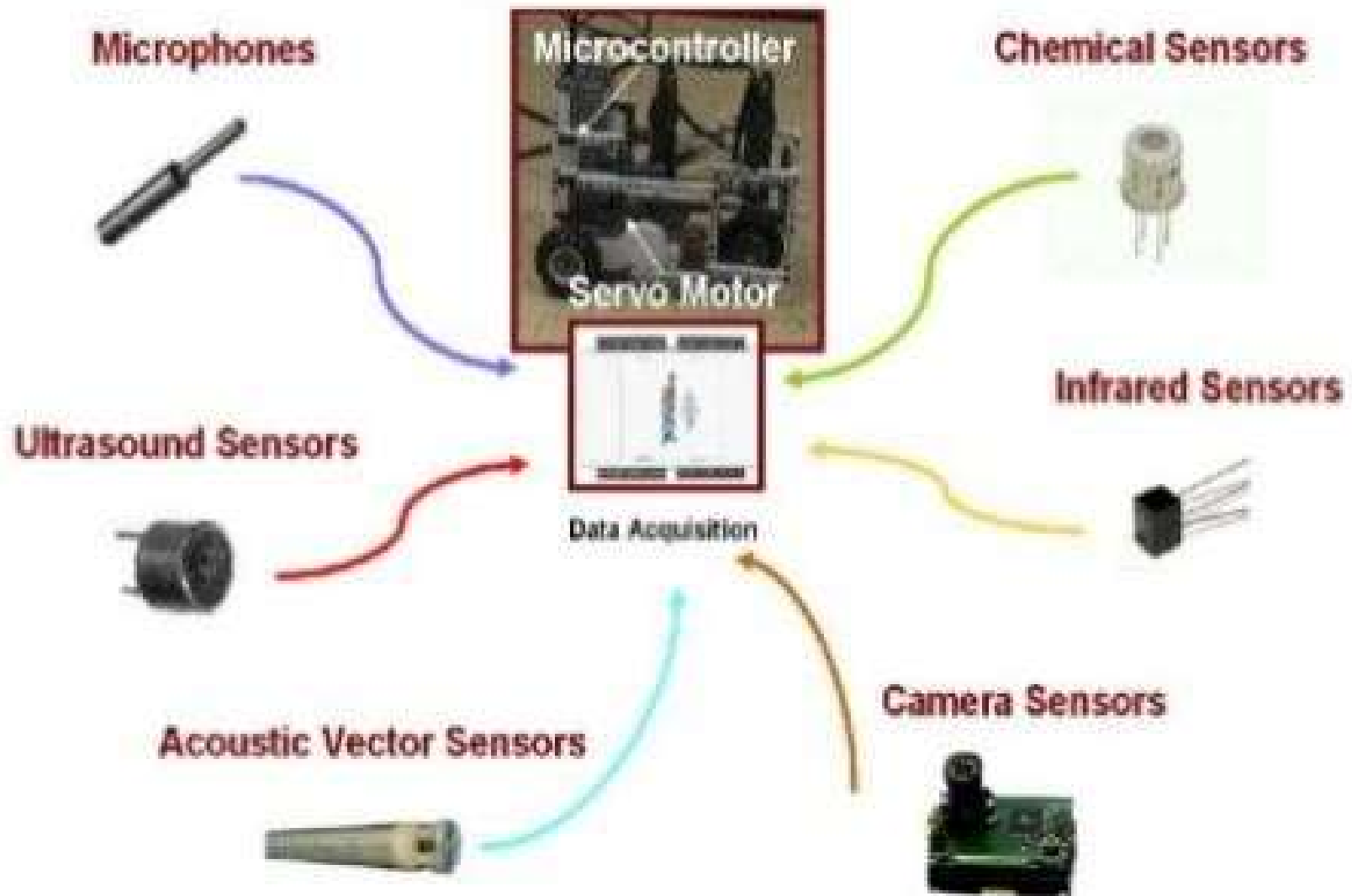
e.g. on/off switches, ultrasonic, force sensor, hall effect, inductive sensor, piezo sensor

Sensor Type	Description
Tactile sensors	Used to determine whether contact is made between sensor and another object. Two types: touch sensors which indicate when contact is made and force sensors which indicate the magnitude of the force with the object.
Proximity sensors	Used to determine how close an object is to the sensor. Also called a range sensor.
Optical sensors	Photocells and other photometric devices that are used to detect the presence or absence of objects. Often used in conjunction to proximity sensors.

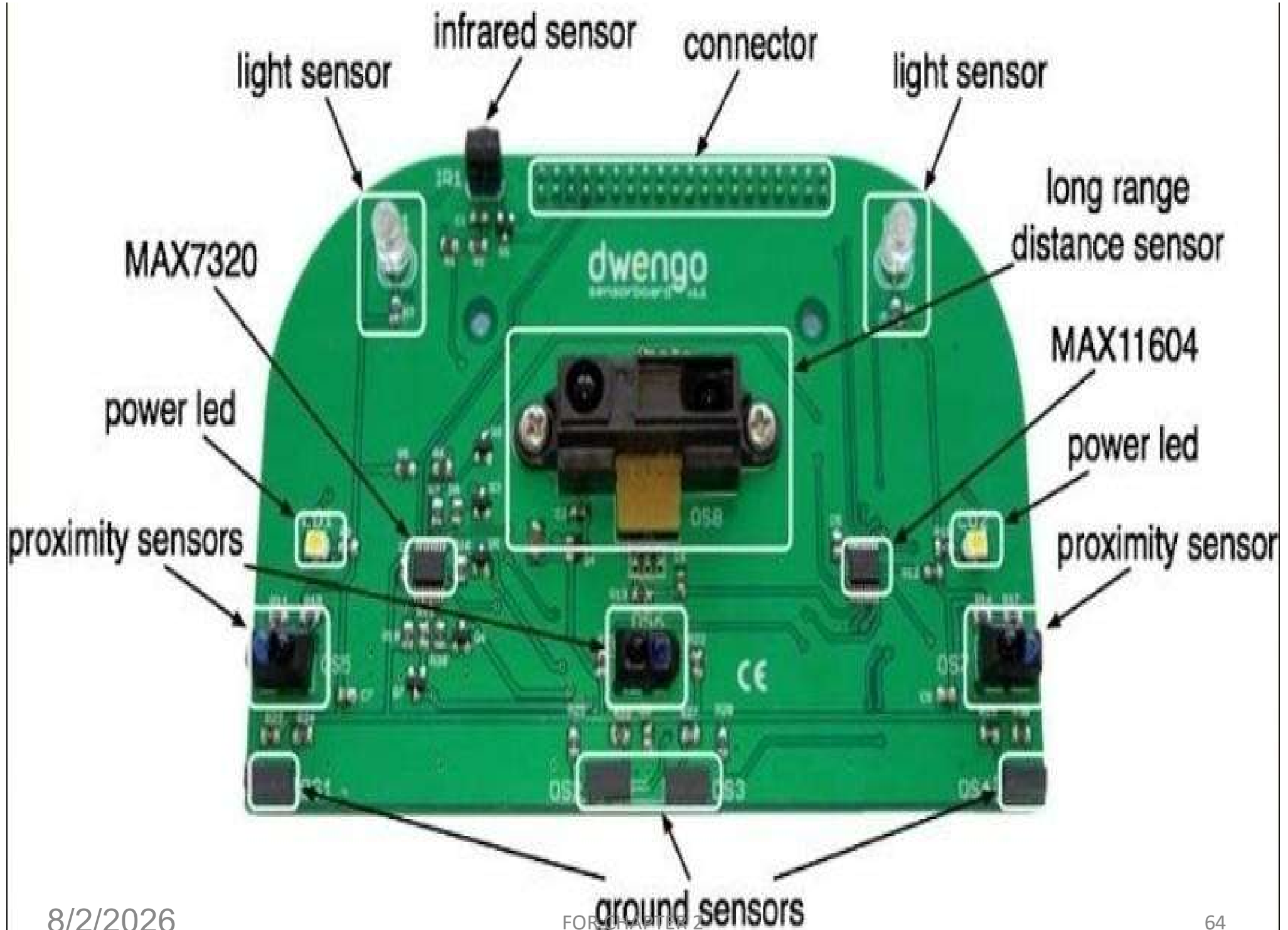
Sensor Type	Description
Machine vision	Used in robotics for inspection, parts identification, guidance and other uses.
Miscellaneous category	temperature, fluid pressure, fluid flow, electrical voltage, current and other physical properties.

SENSORS USED IN ROBOTICS

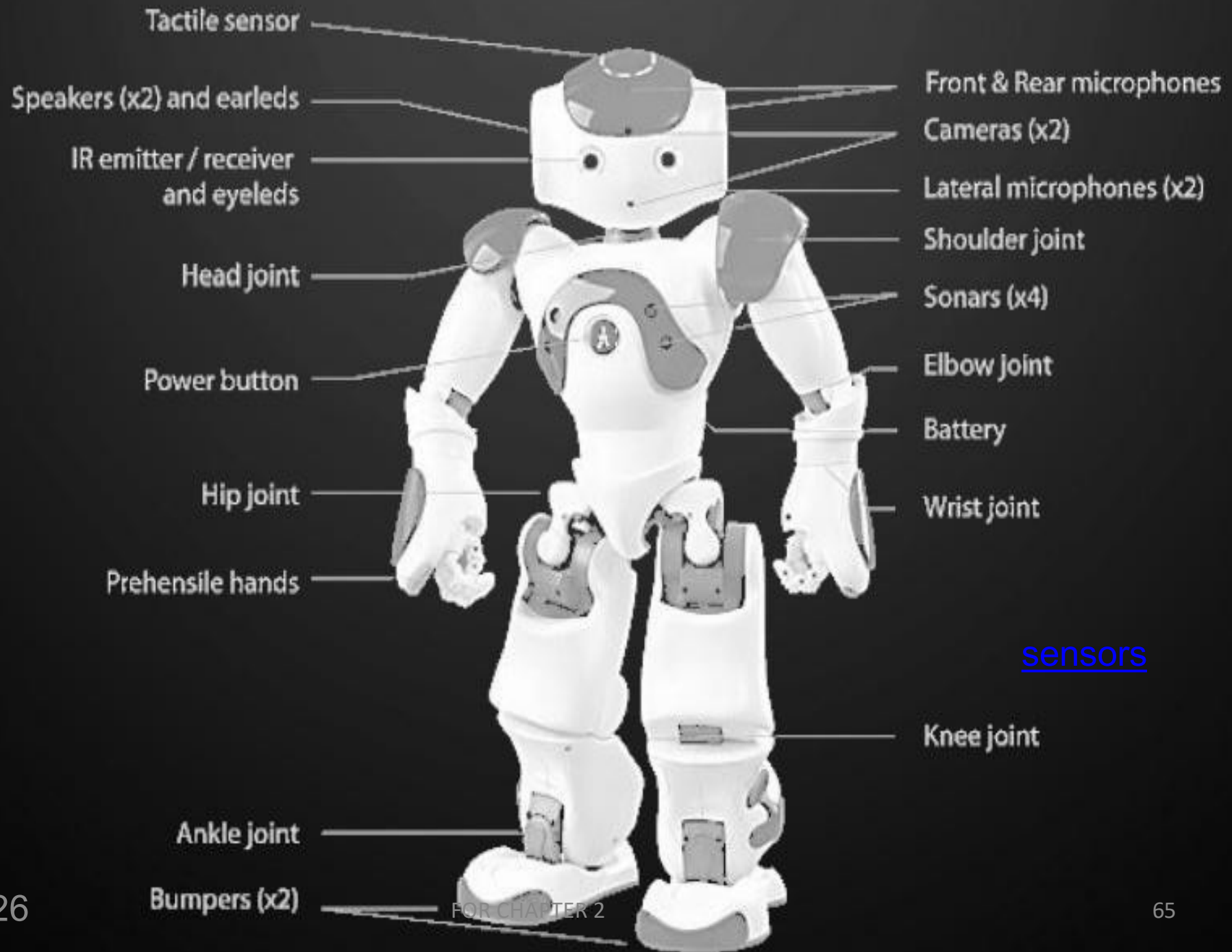
- **a) Proximity Sensor:** This type of sensor is capable of pointing out the availability of a component. Generally, the proximity sensor will be placed in the robot moving part such as end effector. This sensor will be turned ON at a specified distance, which will be measured by means of feet or millimeters. It is also used to find the presence of a human being in the work volume so that the accidents can be reduced.
- **Infrared (IR) Transceivers:** An IR LED transmits a beam of IR light and if it finds an obstacle, the light is simply reflected back which is captured by an IR receiver. Few IR transceivers can also be used for distance measurement.
- **Ultrasonic Sensor:** These sensors generate high frequency sound waves; the received echo suggests an object interruption. Ultrasonic Sensors can also be used for distance measurement.
- **Photoresistor:** Photoresistor is a light sensor; but, it can still be used as a proximity sensor. When an object comes in close proximity to the sensor, the amount of light changes which in turn changes the resistance of the Photoresistor. This change can be detected and processed.



SENSORS USED IN ROBOT SENSING



Advanced sensor model technologies for robotics



Proximity Sensing

Proximity sensing normally means detecting:

- a. Presence or absence of an object.
- b. The size or simple shape of an object.

Proximity sensors can be further classified as contact or non - contact, and as analog or digital in operation. The choice of sensor is determined by the physical, environmental and control conditions. They include the following:

Mechanical - Any suitable mechanical / electrical switch may be adopted but because a certain amount of force is required to operate a mechanical switch it is common to use microswitches.

Pneumatic - These proximity sensors operate by breaking or disturbing an air flow. The Pneumatic proximity sensor is an example of a contact type sensor.

Optical - In the simplest form, optical proximity sensors operate by breaking a light beam which falls onto a light sensitive device such as a photocell. These are examples of non contact sensors. Care must be exercised with the lighting environment of these sensors for example optical sensors can be blinded by flashes from arc welding processes, airborne dust and smoke clouds may impede light transmission etc.

Electrical -

Electrical proximity sensors may be contact or non - contact. Simple contact Sensors operate by making the sensor and the component complete an Electrical circuit. Non – contact electrical proximity sensors rely on the electrical principles of either induction for Detecting metals or capacitance for detecting non metals as well.

Range Sensing -

Range sensing detects how near or far a component is from the sensing position, although they can also be used as proximity sensors. Distance or Range sensors use non - contact analog techniques. Short range sensing, between a few millimetres and a few hundred millimetres is carried out using Electrical capacitance, inductance and magnetic technique. Longer range sensing is carried out using transmitted energy waves of various types.

Eg. radio waves, sound waves and lasers.

Position sensors measure the **position of a joint** (the degree to which the joint is extended). They include:

Encoder: a digital optical device that converts motion into a sequence of digital pulses.

Potentiometer: a variable resistance device that expresses linear or angular displacements in terms of voltage.

Linear variable differential transformer: a displacement transducer that provides high accuracy. It generates an AC signal whose magnitude is a function of the displacement of a moving core.

A **velocity or speed sensor** measures consecutive **position measurements** at known intervals and computes **the time rate of change in the position values**.

Tachometer: A tachometer is fitted to the actuator of the robot to calculate the rotational speed or velocity of shaft.

Laser Surface Velocimeter: It is a non contact optical speed sensor. It uses doppler principle to evaluate the laser light scattered back from a moving object.

Piezoelectric sensor: Uses piezoelectric effect to measure changes in velocity, pressure, temperature etc..

A **proximity sensor** is a sensor able to detect the presence of nearby objects without any physical contact. This sense keeps a **robot** from not running into things. It can also be used to measure the distance from a **robot** to some object.

Infrared (IR) Transceivers: An IR LED transmits a beam of IR light and if it finds an obstacle, the light is simply reflected back which is captured by an IR receiver. Few IR transceivers can also be used for distance measurement.

Ultrasonic Sensor: These sensors generate high frequency sound waves; the received echo suggests an object interruption. Ultrasonic Sensors can also be used for distance measurement.

Photoresistor: Photoresistor is a light sensor; but, it can still be used as a proximity sensor. When an object comes in close proximity to the sensor, the amount of light changes which in turn changes the resistance of the Photoresistor. This change can be detected and processed.

Force Sensing

There are six types of forces (as shown below) that may require sensing. In each case the application of the force may be static (Stationary) or dynamic t Moving). Force is a vector quantity in that it must be specified in both magnitude and direction. Force sensors are therefore analog an operation and sensitive to the direction in which they act.

- Tensile Force
- Compressive Force
- Shear force
- Torsional Force
- Bending Force
- Frictional Force

A number of techniques exist for sensing force, some direct and some indirect.

Tensile Forces: -Can be determined by Strain Gauges, these show a change in their electrical resistance when their length is increased. These gauges measure change in electrical resistance which can be translated into force and are therefore indirect devices.

Compressive Forces: -

Can be determined by devices known as Load Cells, these operate by detecting either a change in dimension of the cell under compressive load or by detecting an increase in the pressure within the cell under load or by exhibiting a change in electrical resistance under a compressive load.

Torsional and Bending Forces: -

Can be regarded as a combination of tensile and compressive forces so a combination of the above technique are employed.

Frictional Forces: -

These relate to situations where movement is to be restrained, so friction force is Detected indirectly using a combination of force and movement sensors.

Tactile Sensing

- Tactile sensing means sensing through touch.
- The simplest types of tactile sensors use an array of simple touch sensors arranged in rows and columns.
- These are commonly called matrix sensors.
- Each individual sensor is activated when brought into contact with the object.
- By detecting which sensors are active (digital) or the magnitude of the output signal (analog) an imprint of the component can be determined. The imprint is then compared to previously stored imprint information to determine the size or shape of the component.
- Mechanical, optical and electrical tactile sensors are available.

Touch

- **Signal Processing:** Touch sensory signals can be generated by the robot's own movements. It is important to identify only the external tactile signals for accurate operations. Recent solution applies an adaptive filter to the robot's logic. It enables the robot to predict the resulting sensor signals of its internal motions, screening these false signals out. The new method improves contact detection and reduces false interpretation.
- **Usage :** Touch patterns enable robots to interpret human emotions in interactive applications. Four measurable features—force, contact time, repetition, and contact area change—can effectively categorize touch patterns through the temporal decision tree classifier to account for the time delay and associate them to human emotions with up to 83% accuracy.

TACTILE SENSORS

- Tactile sensing includes any form of sensing which requires physical touching between the sensor and the object to be sense.
- The need for touch or tactile sensors occurs in many robotic applications, from picking oranges to loading machines. Probably the most important application currently is the general problem of locating, identifying, and organizing parts that need to be assembled.
- **Tactile sensor system includes the capability to detect such things as:**
 1. **Presence**
 2. **Part shape, location, orientation, contour examination**
 3. **Contact are pressure and pressure distribution**
 4. **Force magnitude, location, and direction**
 5. **Surface inspection : texture monitoring, joint checking, damage detection**
 6. **Object classification : recognition, discrimination**
 7. **Grasping : verification, error compensation (slip, position orientation)**
 8. **Assembly monitoring**

Vision

- **Method:** The visual sensing system can be based on anything from the traditional camera, sonar, and laser to the new technology radio frequency identification (RFID), which transmits radio signals to a tag on an object that emits back an identification code. All four methods aim for three procedures—sensation, estimation, and matching.
- **Image processing:** Image quality is important in applications that require excellent robotic vision. Robots can gather more accurate information from the resulting improved image.
- **Usage:** Visual sensors help robots to identify the surrounding and take appropriate action. Robots analyze the image of the immediate environment imported from the visual sensor. The result is compared to the ideal intermediate or end image, so that appropriate movement can be determined to reach the intermediate or final goal.

Hearing

- **Signal processing:** Accurate audio sensor requires low internal noise contribution. Traditionally, audio sensors combine acoustical arrays and microphones to reduce internal noise level. Recent solutions combine also piezoelectric devices. These passive devices use the piezoelectric effect to transform force to voltage, so that the vibration that is causing the internal noise could be eliminated. On average, internal noise up to about 7dB can be reduced.
- Robots may interpret strayed noise as speech instructions. Current voice activity detection (VAD) system uses the complex spectrum circle centroid (CSCC) method and a maximum signal-to-noise ratio (SNR) beamformer.
- **Usage**
Robots can perceive our emotion through the way we talk. Acoustic and linguistic features are generally used to characterize emotions.

Range Sensor

- Range Sensor is implemented in the end effector of a robot to calculate the distance between the sensor and a work part. The values for the distance can be given by the workers on visual data. It can evaluate the size of images and analysis of common objects. The range is measured using the Sonar receivers & transmitters or two TV cameras.

Touch Sensor

- The **touch sensor** has got the ability to sense and detect the touching of a sensor and object. Some of the commonly used simple devices as touch sensors are micro – switches, limit switches, etc. If the end effector gets some contact with any solid part, then this sensor will be handy one to stop the movement of the robot. In addition, it can be used as an inspection device, which has a probe to measure the size of a component.



Touch Sensor

The **force sensor** is included for calculating the forces of several functions like the machine loading & unloading, material handling, and so on that are performed by a robot. This sensor will also be a better one in the assembly process for checking the problems. There are several techniques used in this sensor like Joint Sensing, Robot – Wrist Force Sensing, and Tactile Array Sensing.

Temperature Sensor

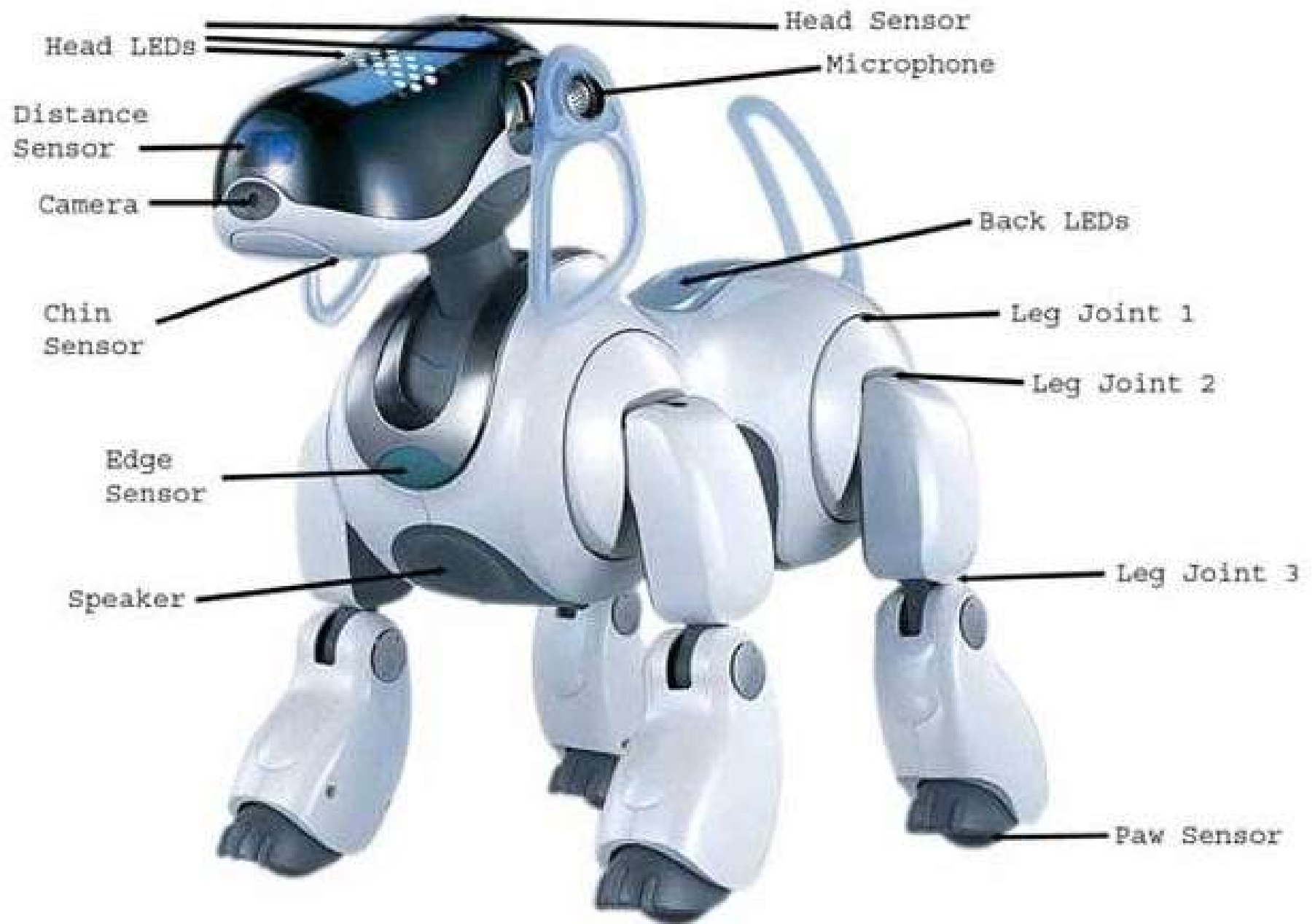
- Tiny temperature sensor ICs provide voltage difference for a change in temperature.
- NTC Thermistor can be assembled in housing in a variety of configurations for temperature sensing, measurement, detection, indicator, monitoring and control.
- Thermistor temperature sensor probe assemblies can conveniently attach to or be an integral part of any system to monitor or control temperature.
- The primary factors which determine the optimum configuration of a thermistor assembly are the operating environment; mounting & time.
- Applications for temperature sensing include air temperature sensors, surface temperature sensors and immersion temperature sensors. Housings for air temperature sensors are often simple, inexpensive devices such as molded plastic shells, deep- drawn brass or aluminum cylinders, or even stainless steel tubes.

SENSORS

- Sensors provide awareness of the environment by sensing things. Sensors are the core of robots. It is the system that alerts the robots..
- Sensing can be in different forms like-

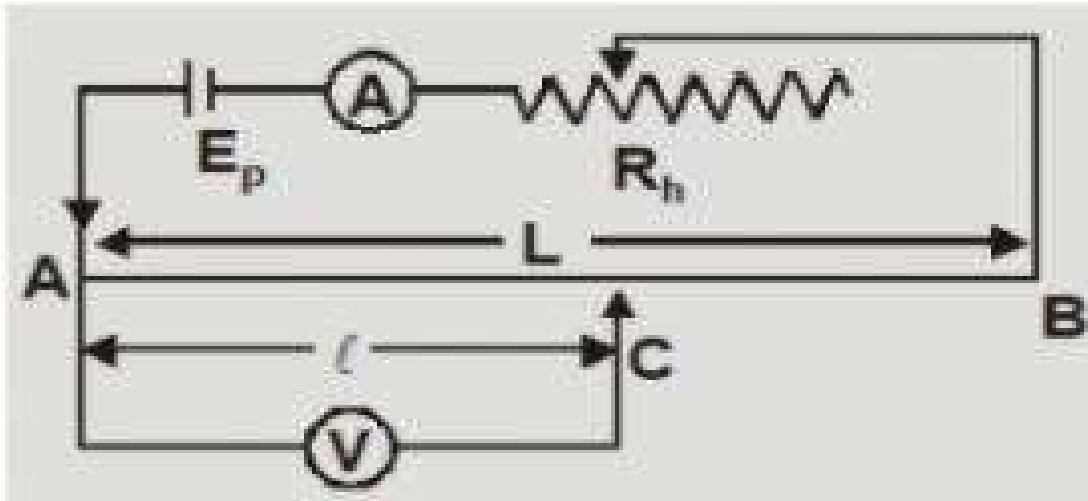
- Light
- Sound
- Heat
- Chemicals
- Force
- Object proximity
- Physical orientation/position
- Magnetic & Electric Fields
- Resistance





INTERNAL SENSORS- POSITION SENSORS

Potentiometer:-



$$V = (E_p / L) l$$

$$\text{or } V = x I;$$

$$\text{where } x = E_p / L$$

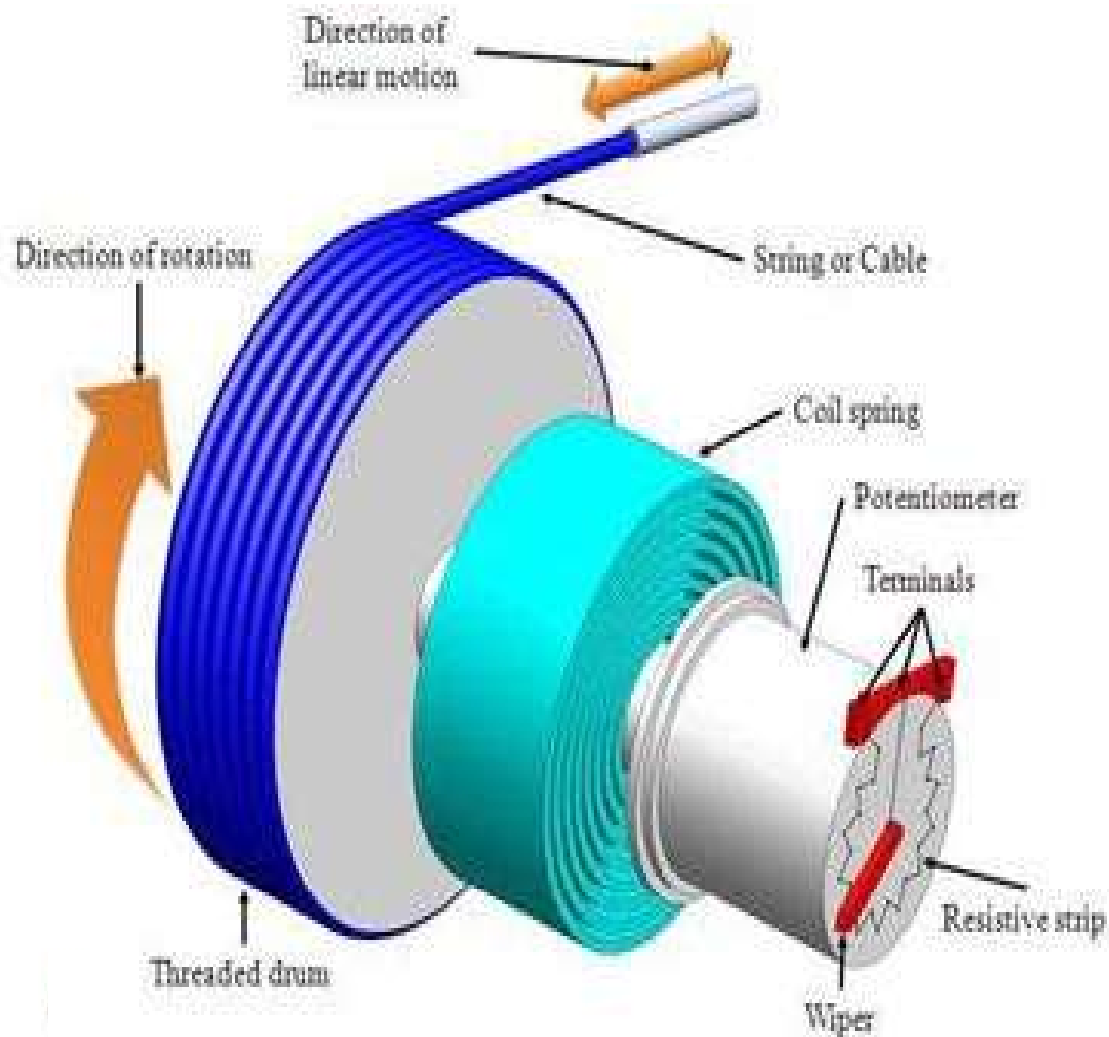
$$\% \text{ resolution} = \left[(V / N) / V \right] \times 100 =$$

$$100 / N$$

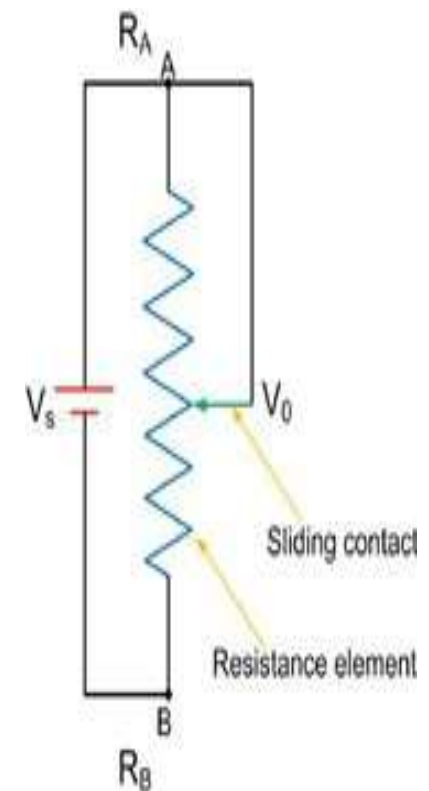
N- number of turns of resistive wire.

A **potentiometer** (also known as a **pot** or **potmeter**) used to measure position is defined as a 3 terminal variable resistor in which the resistance is manually varied to control the flow of electric current. Applied to Robots these devices can be used to measure angular position or linear position of joints. There are two main types of potentiometers: Rotary potentiometer, Linear potentiometer

INTERNAL SENSORS- POSITION SENSORS



Schematic of a potentiometer sensor for measurement of linear displacement



Potentiometer: electric circuit

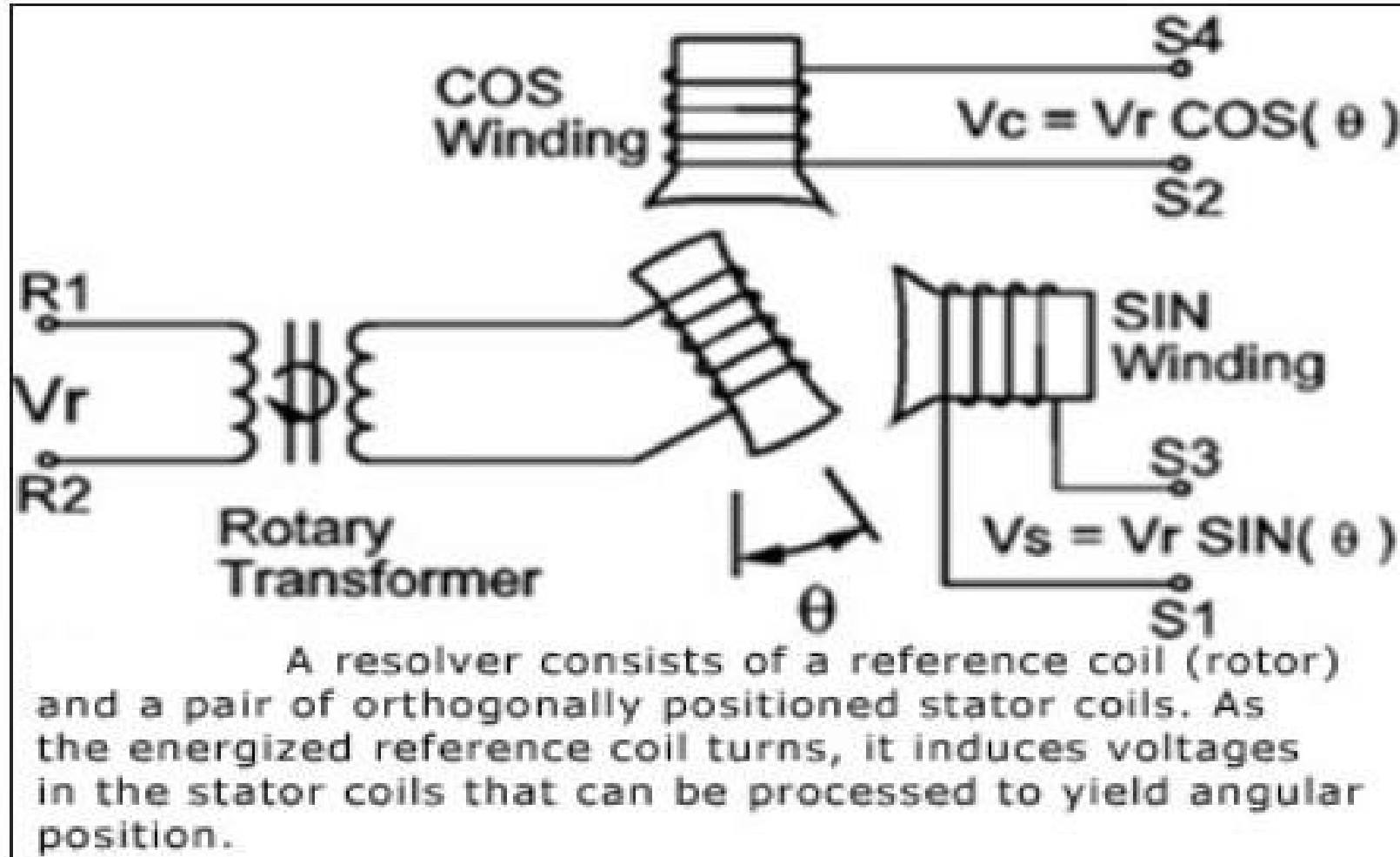
Figure above shows the construction of a **rotary type potentiometer sensor** employed to measure the linear displacement. The potentiometer can be of linear or angular type. It works on the principle of conversion of mechanical displacement into an electrical signal. The sensor has a resistive element and a sliding contact (wiper). The slider moves along this conductive body, acting as a movable electric contact. The object of whose displacement is to be measured is connected to the slider by using • a rotating shaft (for angular displacement) • a moving rod (for linear displacement) • a cable that is kept stretched during operation. During the sensing operation, a voltage V_s is applied across the resistive element. A voltage divider circuit is formed when slider comes into contact with the wire. The output voltage (V_A) is measured as shown in the figure below. The output voltage is proportional to the displacement of the slider over the wire. Then the output parameter displacement is calibrated against the output voltage V_A .

Applications of potentiometer:

- These sensors are primarily used in the control systems with a feedback loop to ensure that the moving member or component reaches its commanded position.
- These are typically used on machine-tool controls, elevators, liquid-level assemblies, forklift trucks, automobile throttle controls.
- In manufacturing, these are used in control of injection moulding machines, woodworking machinery, printing, spraying, robotics, etc.
- These are also used in computer-controlled monitoring of sports equipment.

INTERNAL SENSORS- POSITION SENSORS

Resolvers:-



INTERNAL SENSORS- POSITION SENSORS-Resolver

Working: The primary winding of the stator is connected to a high frequency sine signal. This sine signal is transmitted to rotor winding, because the stator's primary winding and the rotor winding act together like a transformer. Furthermore, we can call the rotor winding the Reference winding.

The pulsating alternating magnetic field of the rotor winding now induces an alternating voltage in the measuring windings sine and cosine. Their amplitudes, however, depend on the angular position of the rotor.

If the rotor winding and the measuring winding are parallel to one another, the magnetic rotor field completely passes through the measuring coil and therefore the induced voltage is maximal.

However, if the rotor winding and measuring winding are at right angles to each other, no voltage is induced.

Resolvers are designed as rotary position sensors, able to monitor a motor or actuator shaft for angular position, direction, and speed. They are analog devices with no onboard electronics, which makes them both simple and robust. The basic design consists of a reference coil (rotor) coupled to the shaft being monitored and a pair of stationary stator coils positioned orthogonal to one another and designated as the **sine** and **cosine** coils.

If we excite the rotor coil with an AC reference voltage V_r given by:

$$V_r(t) = V_0 \sin \omega_r t$$

where ω_r is excitation frequency, the reference voltage inductively couples to the stator coils. This induces the corresponding sine and cosine voltages $V_s(t)$ and $V_c(t)$, respectively:

$$V_s(t) = k V_r(t) \sin \theta = k V_0 \sin \omega_r t \sin \theta$$

$$V_c(t) = k V_r(t) \cos \theta = k V_0 \sin \omega_r t \cos \theta$$

where k is the transformation ratio of output voltage to input voltage and θ represents the angle of the rotor with respect to the home position. We can solve for θ as:

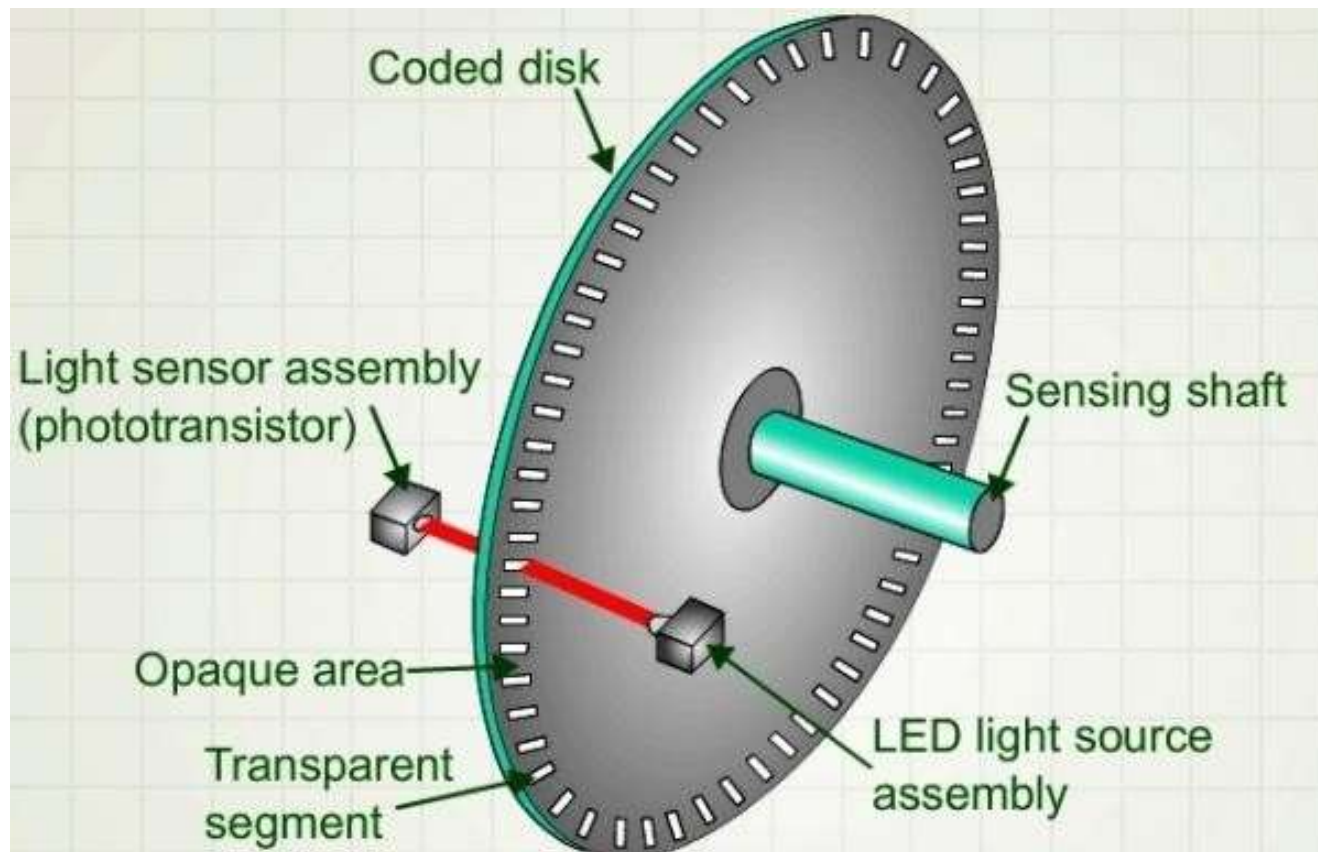
$$\theta = \arctan\left(\frac{\sin \theta}{\cos \theta}\right)$$

$$= \arctan\left[\frac{\frac{V_s(t)}{kV_o(t) \sin \omega_r t}}{\frac{V_c(t)}{kV_o(t) \sin \omega_r t}}\right]$$

$$= \arctan\left[\frac{V_s(t)}{V_c(t)}\right]$$

INTERNAL SENSORS- POSITION/ VELOCITY SENSORS

ENCODER:-



INTERNAL SENSORS- POSITION/ VELOCITY SENSORS

- The **encoder** is an electromechanical device that can measure displacement and velocity (eg. joint position and velocity of robot)
- Encoders are normally digital displacement transducers, consisting of a mechanical element and a sensing head, typically of optical type.
- The mechanical element can be a disc (for rotary type encoders) or a ruler (for linear type encoders) with deposited or carved patterns.
- The sensing head includes a light source (LED) and a light sensor (photo detector) to read the generated code (the encoder output). An absolute encoder has many pulse tracks on its transducer disk.
- When the disk of an absolute encoder rotates, several pulse trains – equal in number to the tracks on the disk – are generated simultaneously.

INTERNAL SENSORS- POSITION/ VELOCITY SENSORS

- At a given instant, the magnitude of each pulse signal will have one of two signal levels (i.e., a binary state) as determined by a level detector. This signal level corresponds to a binary digit (0 or 1). Hence, the set of pulse trains gives an encoded binary number at any instant.

Advantages of encoders:

- High resolution (depending on the word size of the encoder output and the number of pulses per revolution of the encoder)
- High accuracy (particularly due to noise immunity of digital signals and superior construction)
- Relative ease of adaptation in digital control systems (because transducer output is digital)

INTERNAL SENSORS- POSITION/ VELOCITY SENSORS

Applications of Encoders:

- Encoders for the Electronics Industry
- Encoders for CNC Machining
- Encoders for Automation
- Encoders in the Medical Industry
- Encoders in Robotics

Whether it's an articulating arm that picks and places equipment in a manufacturing plant or more mobile, automated, guided robots, they all use encoders.

By their nature, automated systems need these effective speed and positioning feedback systems to function with limited human assistance.

In many cases, encoders with a small profile are best suited for a robotic design, so that all the required technology can fit together in a robot that's sized appropriately to its use.

INTERNAL SENSORS- ACCELERATION SENSORS

ACCELEROMETER

It's an electromechanical device that measures **acceleration**, **vibration** and **tilt angles** by using the MEMS (Micro Electro Mechanical Systems) technology, where acceleration is the rate of change of velocity with respect to the time. Acceleration forces may be Static or Dynamic depending upon the way one uses the accelerometer. The change in the acceleration leads to change in the voltage output of the accelerometer.

Types of accelerometers:

HALL EFFECT ACCELEROMETERS:

The alternating magnetic fields are sensed with respect to different motions and then the corresponding motion is converted into an electric signal.

PIEZO ELECTRIC ACCELEROMETERS:

The piezoelectric crystals have the property of generating voltages when they are pressurized. This property is induced in these type of accelerometers to sense the motion and produce corresponding signals.

INTERNAL SENSORS- ACCELERATION SENSORS

CAPACITIVE ACCELEROMETERS:

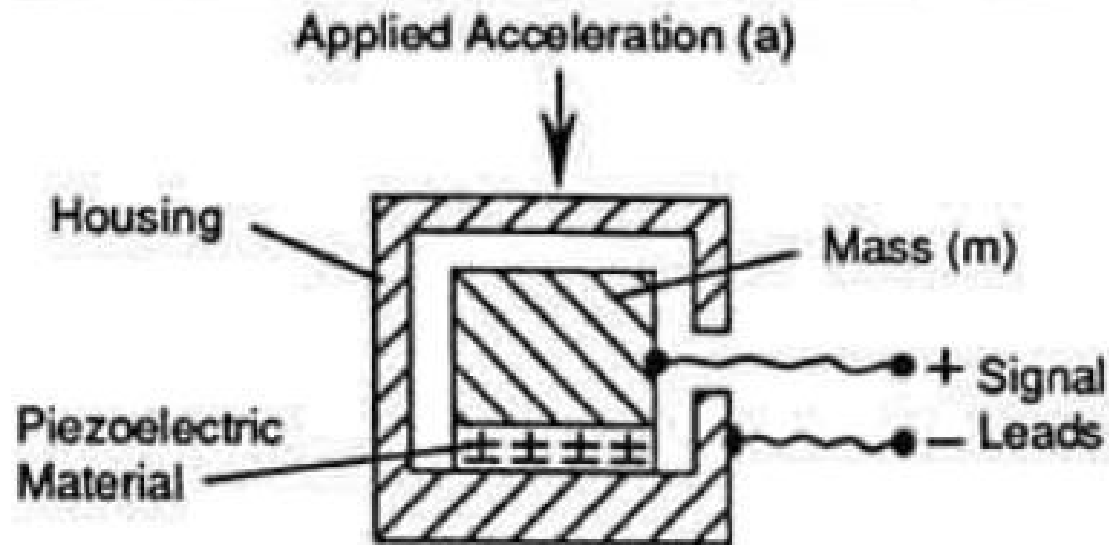
The two micro structures placed nearby, induces a capacitive value between them whose capacitance value changes according to the motion. So by converting the capacitance value into voltage, the motion can be sensed.

USES OF ACCELEROMETERS:

Accelerometers have been extensively used in the field of Automation and Robotics. For example, they are embedded inside the latest versions of laptops (Apple) for preventing the hard disk being damaged by unexpected free fall of them. So basically in these laptops, the hard disk gets switched off automatically whenever accelerometer senses a free fall.

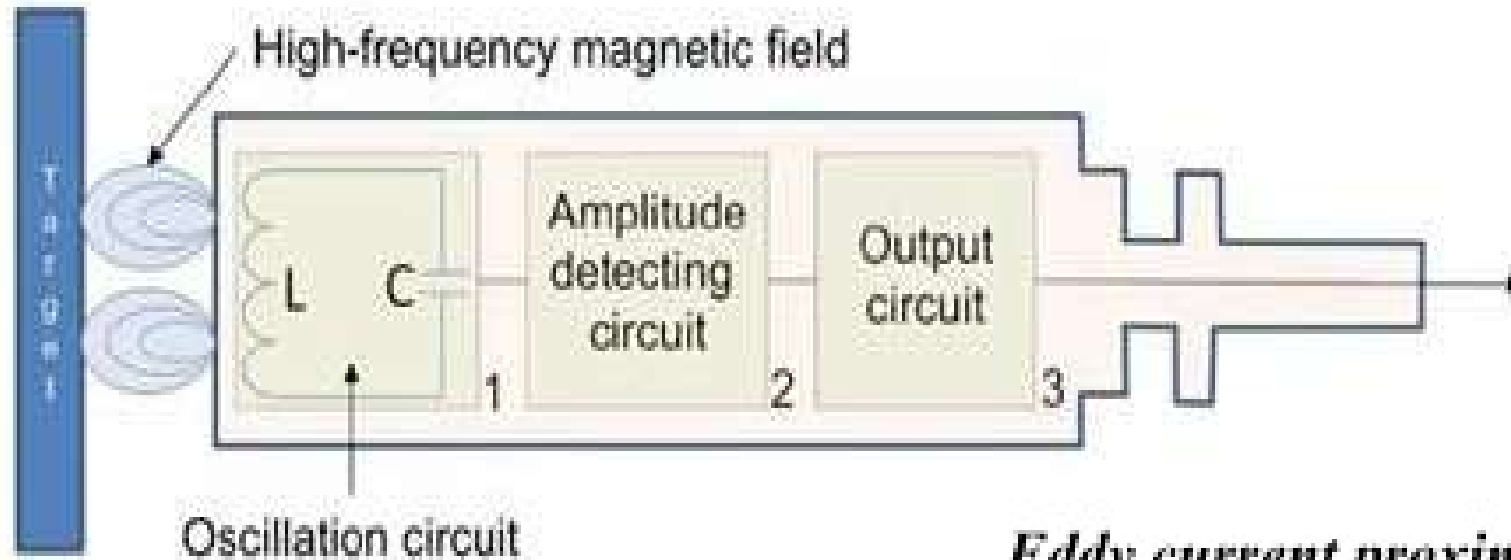
They are also used in Humanoids along with gyro sensors for maintaining the center of gravity positioned in center of the body. Accelerometers detect vibrations and can be used for testing car engine. They can sense the car crashes and push out the airbags at the right time saving the lives of people in the car.

INTERNAL SENSORS- ACCELERATION SENSORS



The main working principle of an accelerometer is that it converts mechanical energy into electrical energy. When a mass is kept on the sensor which is actually just like a spring it starts moving down. Since it is moving down it starts experiencing the acceleration. That acceleration then gets converted into an amount of electric signal which is used for the measurements of variation in the position of the device. The accelerometer can be found with both the forms analog as well as digital form devices.

EXTERNAL SENSOR - INDUCTIVE PROXIMITY SENSOR

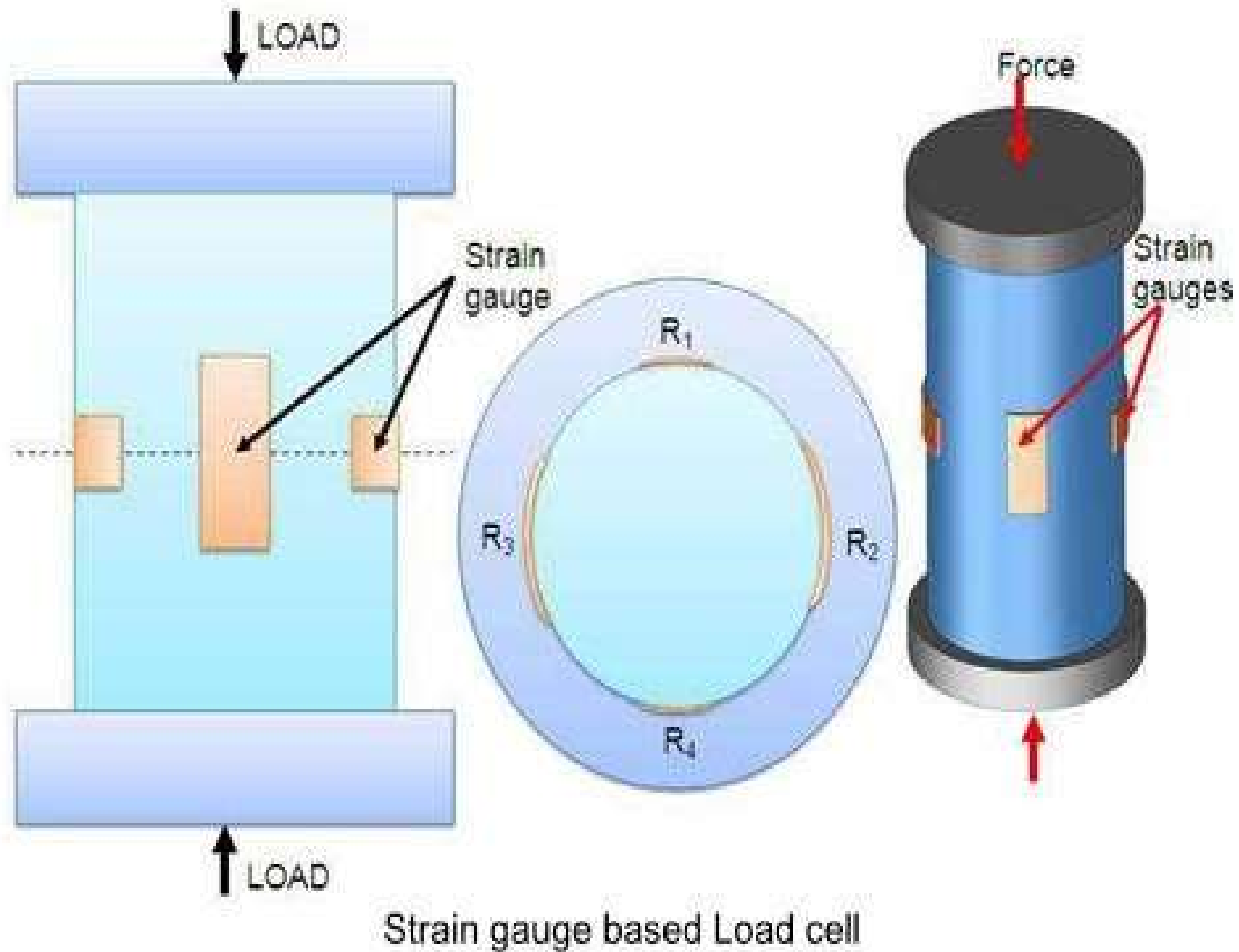


Applications of eddy current proximity sensors

- Automation requiring precise location
- Machine tool monitoring
- Final assembly of precision equipment such as disk drives
- Measuring the dynamics of a continuously moving target, such as a vibrating element.
- Drive shaft monitoring
- Vibration measurements

- **Eddy current proximity sensors** are used to detect non-magnetic but conductive materials.
- They comprise of a coil, an oscillator, a detector and a triggering circuit. Figure shows the construction of eddy current proximity switch.
- When an alternating current is passed thru this coil, an alternative magnetic field is generated. If a metal object comes in the close proximity of the coil, then eddy currents are induced in the object due to the magnetic field.
- These eddy currents create their own magnetic field which distorts the magnetic field responsible for their generation. As a result, impedance of the coil changes and so the amplitude of alternating current.
- This can be used to trigger a switch at some pre-determined level of change in current.
- Eddy current sensors are relatively inexpensive, available in small in size, highly reliable and have high sensitivity for small displacements.

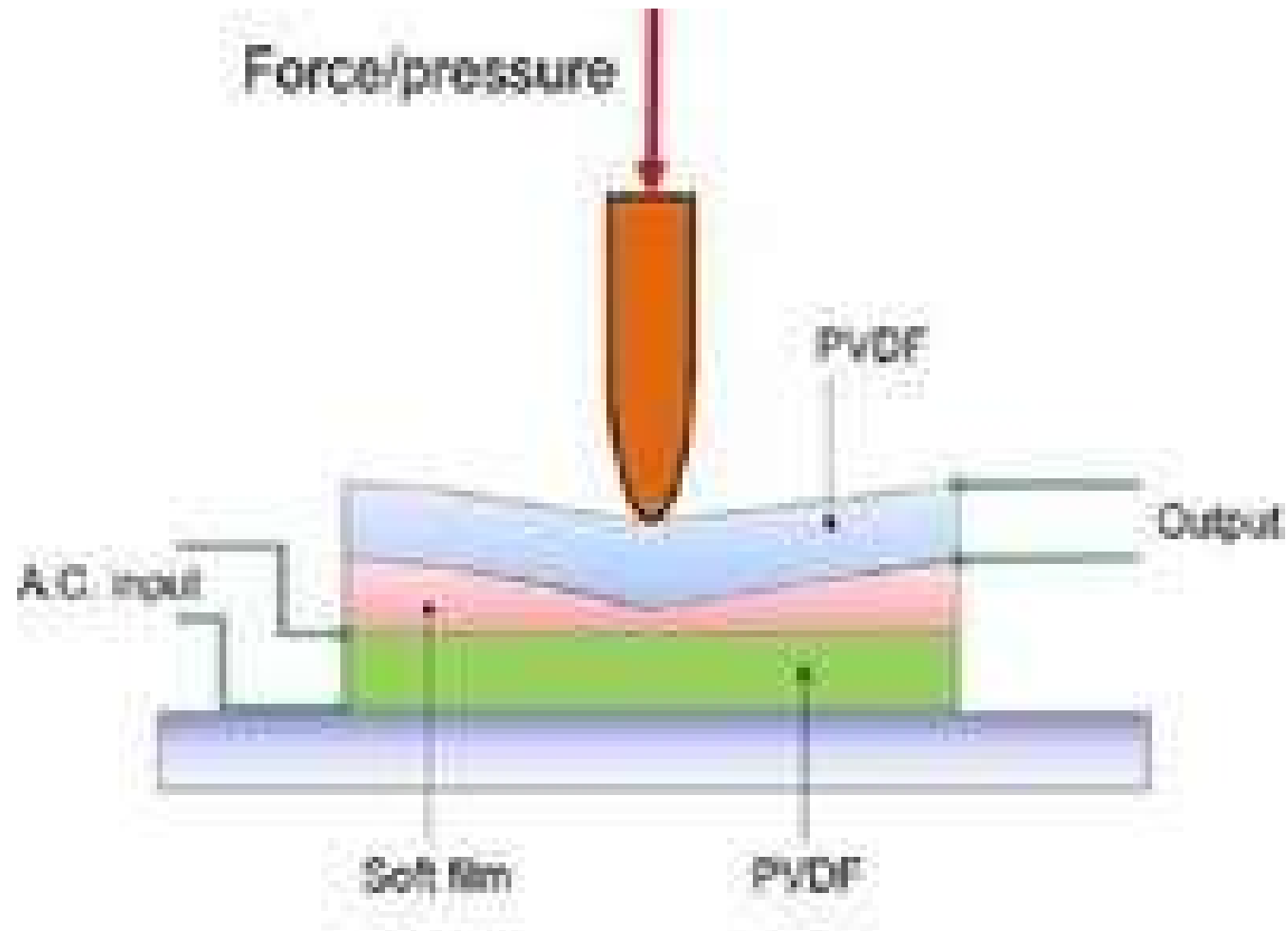
EXTERNAL SENSOR - FORCE SENSOR – STRAIN GAUGE



EXTERNAL SENSOR - FORCE SENSOR – STRAIN GAUGE

Strain gauge based sensors work on the principle of change in electrical resistance. When, a mechanical element subjects to a tension or a compression the electric resistance of the material changes. This is used to measure the force acted upon the element. Figure above shows a strain gauge load cell. It comprises of cylindrical tube to which strain gauges are attached. A load applied on the top collar of the cylinder compress the strain gauge element which changes its electrical resistance. Generally strain gauges are used to measure forces up to 10 MN. The non-linearity and repeatability errors of this transducer are $\pm 0.03\%$ and $\pm 0.02\%$ respectively.

TOUCH/ FORCE SENSOR – TACTILE SENSOR



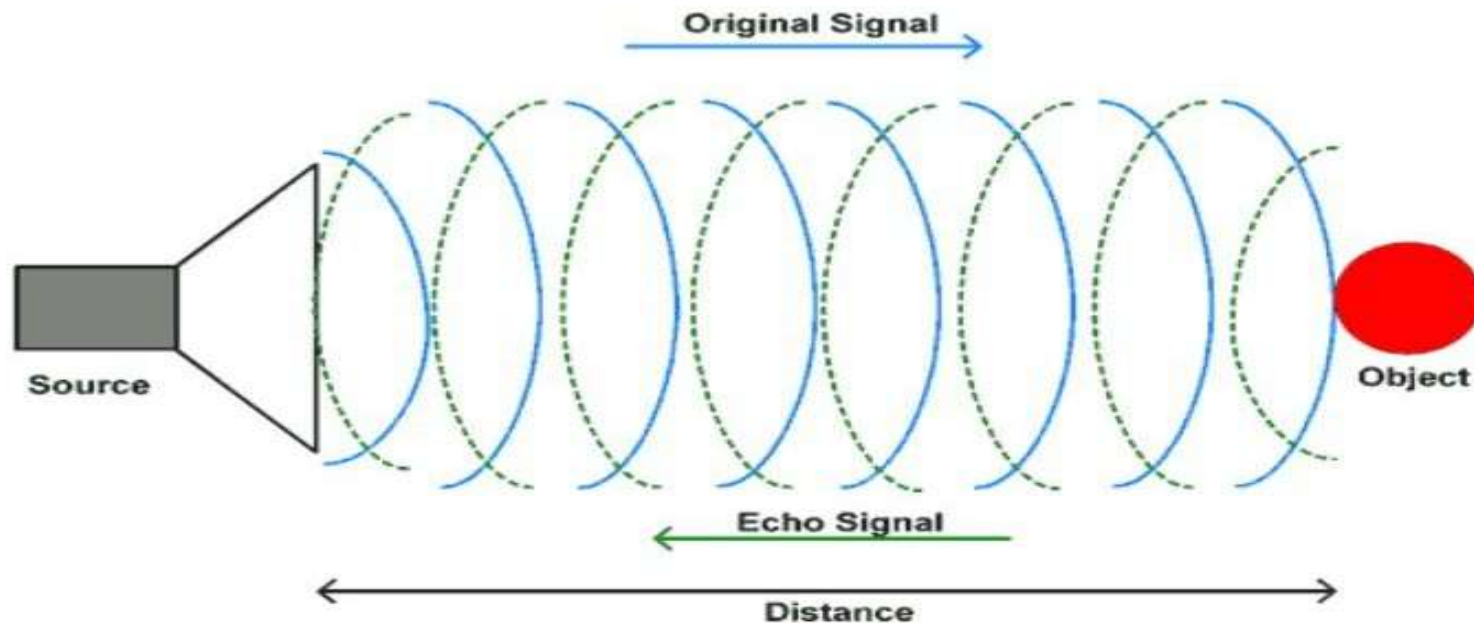
TOUCH/ FORCE SENSOR – TACTILE SENSOR

- Tactile sensors are used to sense the contact of fingertips of a robot with an object. They are also used in manufacturing of 'touch display' screens of visual display units (VDUs) of CNC machine tools.
- Figure shows the construction of piezo-electric polyvinylidene fluoride (PVDF) based tactile sensor. It has two PVDF layers separated by a soft film which transmits the vibrations.
- An alternating current is applied to lower PVDF layer which generates vibrations due to reverse piezoelectric effect. These vibrations are transmitted to the upper PVDF layer via soft film.
- These vibrations cause alternating voltage across the upper PVDF layer. When some pressure is applied on the upper PVDF layer the vibrations gets affected and the output voltage changes.
- This triggers a switch or an action in robots or touch displays.

EXTERNAL SENSOR – RANGE SENSOR – ULTRASONIC SENSOR

The ultrasonic sensor works on the principle of SONAR and RADAR system which is used to determine the distance to an object.

An ultrasonic sensor generates the high-frequency sound (ultrasound) waves. When this ultrasound hits the object, it reflects as echo which is sensed by the receiver as shown in below figure. By measuring the time required for the echo to reach to the receiver, we can calculate the distance. This is the basic working principle of Ultrasonic module to measure distance or range.



EXTERNAL SENSOR – RANGE SENSOR – ULTRASONIC SENSOR

How to calculate distance?

We know that,

$$\text{Distance} = \text{Speed} \times \text{Time}$$

The speed of sound waves is 343 m/s.

So,

$$\text{Total Distance} = \frac{343 \times \text{Time of High(Echo) Pulse}}{2}$$

Total distance is divided by 2 because signal travels from **emitter** to **object** and returns to the **receiver**.